

A High-Performance GaN Power Module With Parallel Packaging for High-Current and Low-Voltage Traction Inverter Applications

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Abstract—Gallium nitride (GaN) power semiconductors are being explored as promising alternatives for the next generation of high-power traction inverters, suitable for both high- and low-voltage applications. In such scenarios, the peak current can reach several hundred amps. Hence, using multiple GaN high-electron-mobility transistor (HEMT) chips or dies in parallel is definitely a design approach to extend power capacity and minimize power losses. However, the parallelization and packaging structure of GaN devices presents numerous challenges. Achieving balanced dynamic and static current sharing between paralleled devices, as well as ensuring robust gate driving signals, is indispensable for reliable operation during high dV/dt and dI/dt transients, particularly with switching noise-sensitive gate-source threshold voltages. In addition, thermal dissipation and mechanical stress must fulfill industrial requirements. This article introduces a compact 100 V/360-A GaN power module specifically designed for packaging on insulated metal substrates (IMSS) or direct-bonded copper (DBC). The proposed packaging approach offers superior electrical performance, excellent thermal dissipation, minimal mechanical stress, and a cost-effective solution. Finally, the advantages of the proposed power module design are validated through various experimental setups, including a 48 V/328-A double pulse test (DPT), a 48/24-V 3-kW buck converter implementation, and a 48 V/5-kW three-phase inverter demonstration.

Index Terms—Gallium nitride high-electron-mobility transistors (GaN HEMTs), high current, paralleling devices, power module packaging.

I. INTRODUCTION

A. Motivation

HIGH-VOLTAGE 400/800-V traction inverters for electric vehicles have been a major focus of academic and industrial research for decades, while light electric

vehicles (LEVs) equipped with 48-V low-voltage batteries are becoming more prevalent in urban areas, thanks to their affordability, safety, and practicality. These traction inverters, designed for low-voltage and high-power applications, generally deliver between 5 and 12 kW of power to the motor. Their input voltage typically falls between 48 and 72 V, resulting in phase-current levels in the range of several hundred amps [1], [2]. Thus, paralleling multiple discrete power devices has long been an indispensable approach to meet the demands of high-performance electric vehicle systems. Over the years, the use of parallel traditional Si MOSFETs has become a well-established and reliable solution for low-voltage motor drives, thanks to their mature technology and reliable operation [2], [3]. However, Si MOSFETs exhibit limitations, such as high ON-state resistance and low switching speed.

In contrast, emergent wide bandgap (WBG) devices, with a focus on gallium nitride high-electron-mobility transistors (GaN-HEMTs), offer significant advantages, including higher switching frequency up to a few MHz [4], lower ON-state resistance, absence of reverse recovery charge, more compact chip size, and superior temperature tolerance [5], [6], [7], [8]. In [9], the study highlights that GaN-based inverters outperform Si-based inverters even in low-frequency scenarios, where motor insulation, bearing current, and filter design concerns limit switching frequency and speed (dV/dt). In comparison to Si MOSFETs, enhancement-mode GaN HEMTs are also considered more suitable for paralleling multiple chips. This suitability is attributed to the gate voltage threshold ($V_{g(th)}$) being less sensitive to junction temperature (T_j) fluctuations [5], [6], [10], [11], [12], along with a positive temperature coefficient and a greater temperature dependence of $R_{DS(on)}$ [11], [13]. However, the parallelization of multiple GaN dies introduces additional challenges, motivating researchers to address them for high-power applications [5], [12], [14], [15], [16], [17], [18], [19], [20], [21], [22]. Many previous studies have primarily focused on power packaging for both nonparallel and parallel configurations of two GaN chips, targeting low-current, and high-voltage applications such as dc-dc converters [5], [7], [18], [23], [24], [25]. These studies have prioritized electrical performance to achieve high switching frequencies, often with less consideration for other essential factors. However, high-power

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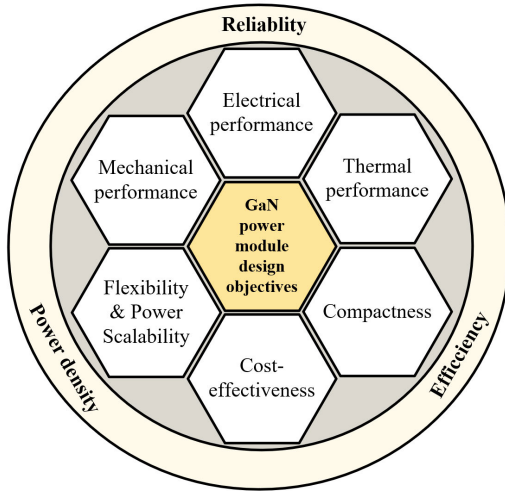


Fig. 1. Multiobjective considerations in high-power GaN power module design.

GaN power module designs require the parallelization of four or more devices, considering multiple design objectives to ensure reliable operation under harsh conditions, as illustrated in Fig. 1.

These key objectives encompass electrical, thermal, and mechanical performance. In addition, cost-effectiveness, flexibility, and power scalability are essential to making these design solutions practical and applicable in the automotive industry. Meeting all design objectives for a GaN-based power module presents significant challenges. In most failure scenarios of power module design, high overshoot voltage, unbalanced current sharing among parallel dies, and unstable gate–source voltage during transient periods are recognized as the main causes of power module destruction. Due to the rapid changes in current (di/dt) and voltage (dV/dt) during switching transients, the negative impact of parasitic components becomes significantly more severe in the packaging of multiple paralleled GaN devices [12], [21]. The asymmetrical arrangement of components and printed circuit board (PCB) layout can lead to dynamic current imbalances, especially during switching transients with currents exceeding hundreds of amperes. Moreover, the GaN switch’s gate–source voltage is highly influenced by the parasitic inductances and capacitances present in the gate loop design. Unlike SiC or Si devices, which have gate voltages that can tolerate ± 20 V and threshold voltages of more than 4.5 V [26], GaN transistors typically require lower gate drive voltages, which can vary depending on specific GaN technologies such as GaN cascode, GaN gate injection transistor (GIT), or e-mode GaN HEMTs [27]. E-mode GaN from GaN Systems, for instance, requires substantially lower gate drive voltages at $+7/-10$ V, with a threshold value of 1.3 V [6]. Similarly, EPC’s enhancement mode gallium nitride (eGaN) devices have a maximum gate voltage of $+6/-5$ V. Consequently, if the power loop and gate driver loop are not meticulously designed, the switches may have a high risk of being unintentionally activated. Moreover, concentrated power losses in compact GaN devices require effective thermal management solutions. Due to the aforementioned challenges, packaging techniques developed for paralleling Si/SiC dies cannot be directly applied to GaN devices.

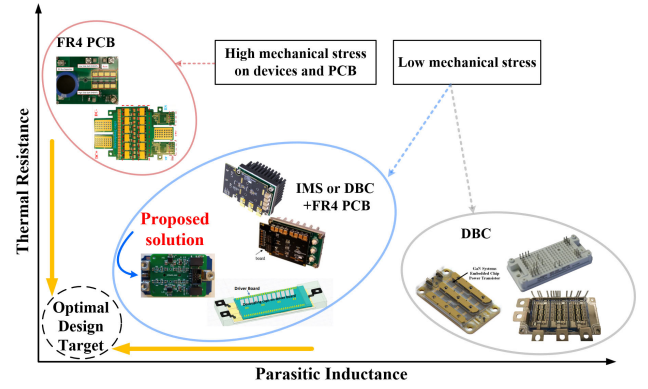


Fig. 2. Brief comparison of different packaging technologies for paralleling GaN devices.

B. State-of-the-Art Overview

Existing literature covers various GaN packaging techniques for high-current applications in both commercial products and academic research. The majority of previous studies have been conducted regarding the parallel configuration of 650-V GaN devices [5]. Nevertheless, there has been comparatively less research attention on the application of low-voltage 100-V GaN for motor drives. Table 1 presents a comparison between previous studies and our proposed design in terms of voltage and current rating, testing results, packaging structure, power loop inductance, thermal performance, complexity, and cost-effectiveness for 400- and 48-V applications. The packaging methods for GaN power modules are generally classified into three groups, as depicted in Fig. 2. The first group involves the use of direct-bonded copper (DBC) substrates to attach GaN chips, with aluminum bond wires providing interconnections. A typical DBC substrate consists of a copper top layer for electrical connectivity, a ceramic middle layer for electrical isolation and heat spreading, and a copper bottom layer soldered to a metal baseplate for mechanical support and heat dissipation. Ceramic materials using Al_2O_3 and AlN are frequently chosen for their superior electrical isolation and thermal conductivity properties, with Al_2O_3 exhibiting thermal conductivity between 20 and 30 W/mK and AlN ranging from 120 to 180 W/mK. Hence, this packaging method can withstand significant thermal and mechanical stresses, making it ideal for high-power applications reaching up to several hundred kilowatts [28]. Nevertheless, the design options of DBC for minimizing and equalizing parasitic inductance in power loops are quite limited, which leads to inferior switching performance [15], [21]. An alternative approach is to adopt FR4-PCBs that provide greater flexibility in the copper trace layout [12], [22]. This allows for the effective integration of magnetic flux cancellation techniques in the multilayer PCB structures to reduce the impact of stray inductance. Thermal vias establish connections between the layers, enabling the transfer of heat from surface-mounted devices to alternate sides of the PCB. Nevertheless, PCBs made of FR4 material still exhibit notably low thermal conductivity values, ranging from 0.2 to 0.3 W/mK. Furthermore, paralleling GaN chips on an FR4-PCB presents the issue of nonuniform dispersion of thermomechanical stress across individual dies or chips,

TABLE I
COMPARISON OF PARALLEL GaN DEVICE STRUCTURES FOR HIGH-CURRENT HALF-BRIDGE POWER MODULE PACKAGING

Reference /Year	Design Voltage/Current Experimental setups	Numbers of paralleled devices and PCB structure	Switching Performance / Power loop inductance	Thermal Performance / Mechanical stress on devices	Complexity / Cost
400V applications					
[15] 2015	• Rated: 650 V/150 A • Tested: 350 V/100 A (DPT)	• 5×GS66508B • a DBC + Gate driver PCB	Poor (5.6 nH)	Best No	Moderate High
[12] 2017	• Rated: 650 V/240 A • Tested: 400 V/24 A (DPT)	• 4× GS66516T • 6-layer FR4 PCB	Very good (0.7 nH)	Moderate Yes	Moderate High
[16] 2018	• Rated: 650 V/240A • Tested: 400 V/136 A (DPT)	• 4× GS66516B • a single layer IMS & a 4-layer FR4	Good (N/A)	Good No	High High
[17] 2017	• Rated: 650V/120A • Tested: 400V/60A (DPT)	• 4×GS66508B • a single layer IMS & a 6-layer FR4	Good (1.3 nH)	Good No	High Moderate
[18] 2021	• Rated: 650 V/120 A • Tested: N/A	• 4×GS66508T • 2 DBC & 2 FR4 PCB	N/A (2.1 nH)	Best No	High Very high
[19] 2021	• Rated: 650 V/350 A • Tested: 450 V/150 A (DPT)	• 2×GS-065-150-1-D 2 • 1 DBC & 1 double layer FR4 PCB	Good (1.54 nH)	Best Yes	High Very High
[20] 2022	• Rated: 650 V/720 A • Tested: 400 V/600 A (DPT)	• 12× GS66516T • 6-layer FR4 PCB	Best (0.52 nH)	Moderate Yes	High Very High
[29] 2020	• Rated: 650 V/1200 A • Tested: 400 V/48 A (DPT)	• 8×GS-065-150-1-D 2 • 1 DBC & 1 double layer FR4 PCB	Moderate (3.37 nH)	Best No	High High
48V applications					
[21] 2019 Commercial	• Rated: 100 V/810 A • Tested: 48 V/275 A (DPT)	• 9× GS61008T • a DBC & Gate driver PCB	Poor Test failure (> 10nH)	Good No	Moderate High
[21] 2019 Customized	• Rated: 100 V/360 A • Tested: 48 V/370 A (DPT)	• (2× GS61008T) ×2 • a 2-layer PCB & a 6-layer FR4 PCB	Good (<1nH)	Moderate Yes	Moderate High
This work	• Rated: 100 V/360 A • Tested: + 48 V/328 A (DPT) – 48 V/400A + 3 kW-48 V/24 V Buck converter + 5 kW- three phase inverter	• 4× GS61008P • a single layer IMS & a 2-layer FR4 PCB	Good (1.14 nH-simulated) (1.17 nH-validated)	Good No	Low Low

resulting in uneven temperature distribution within the parallel structure. Hence, this packaging approach is suitable for a small number of parallel devices and low-power applications. By increasing the number of layers in multilayer PCB structures, the bottleneck of PCB in high-current applications can be improved. However, it will significantly raise the cost of fabrication. The third approach for packaging a high-current GaN power module involves using an insulated metal substrate (IMS), typically made of a copper layer, a dielectric layer, and an aluminum or copper base designed for efficient heat exchange. Its thermal conductivity varies between 2 and 7 W/mK, corresponding to the thermal characteristics of the metal base and the dielectric materials. While IMS has greater thermal resistance compared to DBC, it offers significantly higher thermal conductivity than FR4-based PCBs. In order to increase the design flexibility of the IMS for achieving low parasitic inductance, the IMS can be integrated with the FR4-PCB of the gate driving signal and decoupling capacitor, creating an IMS hybrid structure [5], [16], [17]. Similarly, DBC hybrid structure is the combination of DBC substrate and FR4-PCB [19], [23], [29]. Nevertheless, the power loop inductance in this method has not been minimized due to the thickness of the FR4-PCB and the distance between the PCB and the IMS or DBC power board.

As illustrated in Fig. 2, the hybrid configuration is generally a compromise between PCB and DBC solutions, offering a

balance between switching and thermal performance, making it an attractive solution for medium-power applications.

Regarding the PCB layout structures of GaN devices, the lateral and vertical configurations are the two most common layout methods employed in power module designs. In the lateral structure, the decoupling capacitors and GaN devices are placed on the same side of the PCB, with the output switching node connected to either an internal or an alternate side layer through vias. This conventional lateral half-bridge layout offers a low-cost and straightforward circuit solution. However, it poses challenges in achieving a magnetic field cancellation effect. To reduce the power loop inductance in this layout, an adjacent inner layer can be added for shielding. The induced current in the shield layer, which flows in the opposite direction to the current in the power loop, creates a flux cancellation effect, thereby decreasing the parasitic inductance in the power commutation loop [24], [25]. Nevertheless, achieving identical power loop inductances becomes unattainable when dealing with applications involving the parallelization of multiple devices. In the vertical layout structures, the decoupling capacitors are arranged on either the same side or the opposite side of the GaN switches. The vertical commutation loop is perpendicular to the board plane, with the traces separated into different layers of the PCB. This method leverages the mutual cancellation of magnetic flux to considerably minimize parasitic inductance. In this layout approach, 4–6 layers are

typically required, and the power loop inductance is significantly influenced by the thickness between the layers. In addition, the symmetrical power loop layout can be easier to arrange when paralleling multiple devices. Asymmetrical layout and variations in device parameters, such as ON-state resistance $R_{DS(on)}$ and threshold voltage V_{th} , are identified as the primary factors contributing to dynamic and static current imbalances among the devices [30]. The static current imbalance can be mitigated by the self-balancing mechanisms inherent in the positive temperature coefficient of $R_{DS(on)}$. However, the influence of the temperature dependence of V_{th} and manufacturing threshold voltage deviation on current imbalance is not within the scope of this study. Our initial design concept, as partially presented in [31], included limited analysis, design guidelines, and testing results. This article substantially extends that research by introducing an enhanced hybrid packaging configuration aimed at further reducing power loop inductance and improving current-carrying capability. It provides comprehensive design guidelines, extensive electrothermal simulation results, and experimental validations of both dynamic and continuous electrical operation for the proposed design.

C. Contributions of This Study

This article introduces a novel hybrid packaging configuration that incorporates an IMS substrate with multiple parallel GaN switches and copper bars, all evenly distributed and soldered together, along with decoupling capacitor boards. The power board is then stacked with an FR4-driver board to assemble a 100 V/360-A GaN power module. The aim is to meet various design objectives for high-power GaN modules that are adaptable for both 100- and 650-V bottom-cooled chips, thereby unlocking the potential of GaN devices for different electric vehicle (EV) drivetrain systems. The key features of the proposed GaN power module are outlined below.

- 1) The even distribution of decoupling capacitors on the PCBs across very thin copper bars leads to a symmetrical design for the power loops, ensuring uniform and low parasitic inductances.
- 2) The proposed hybrid IMS structure enables effective heat dissipation without compromising low power loop inductances. In addition, this packaging concept effortlessly adapts to DBC substrates, enhancing thermal performance without modification.
- 3) Parallel GaN chips experience minimal mechanical stress, reducing thermal differences caused by imbalanced pressure from heat sink attachment.
- 4) Offering a cost-effective and simple packaging approach with assured reliable switching operation, the power module maintains a safe margin for V_{DS} overshoot and low-ringing V_{GS} waveforms under high switching currents.
- 5) Featuring a similar input and output terminal arrangement as common industrial power modules, it supports convenient power extension through modular configurations, making it ideal for a wide range of high-power motor drive systems.

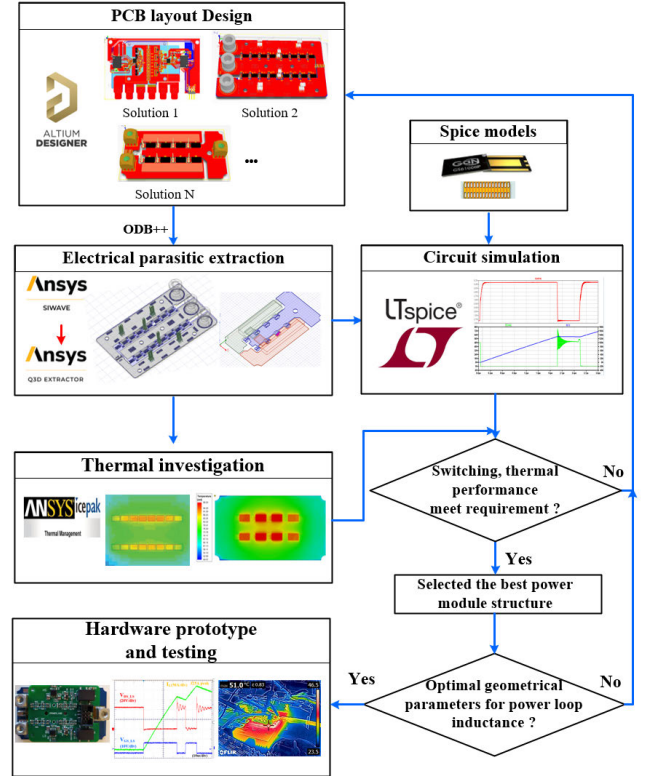


Fig. 3. Digital framework and design process for developing the proposed GaN module.

D. Paper Organization

The rest of the article is structured as follows. Section II introduces the overall digital design framework and package structure of the proposed GaN half-bridge module, including design considerations and feasible solutions. Section III identifies the critical parameters that cause failures in power modules and presents the design of the gate driver. In Section IV, parasitic components are extracted using Ansys Q3D finite element analysis (FEA) software. Subsequently, the switching and thermal performance of the proposed design are validated using LTSPICE and Ansys Icepak software tools. Section V explains the fabrication process of the power module. In Section VI, hardware validation of the power module is conducted through several experimental setups, including a double pulse test (DPT), buck converter, and three-phase inverter. Section VII discusses the results and future perspectives. Finally, Section VIII concludes the article by summarizing key findings.

II. DESIGN STRATEGIES FOR THE PROPOSED GAN POWER MODULE

A. Design Strategies

To analyze and investigate the electrical and thermal behavior of power module packaging solutions, analytical approaches using high-order differential equations have become impractical due to the intricate multiphysics interactions involved. As depicted in Fig. 3, an electrothermal finite element simulation framework is employed to reduce

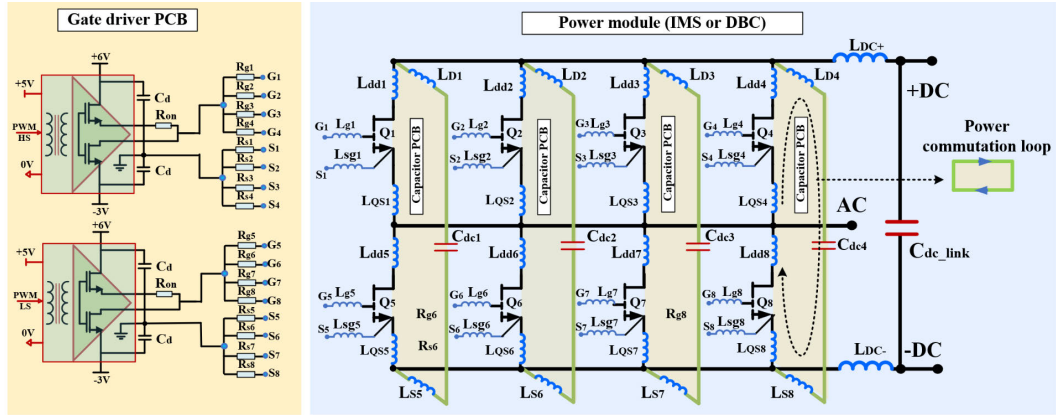


Fig. 4. Circuit diagram of half bridge with four GaN chips in parallel with key packaging parasitic inductances.

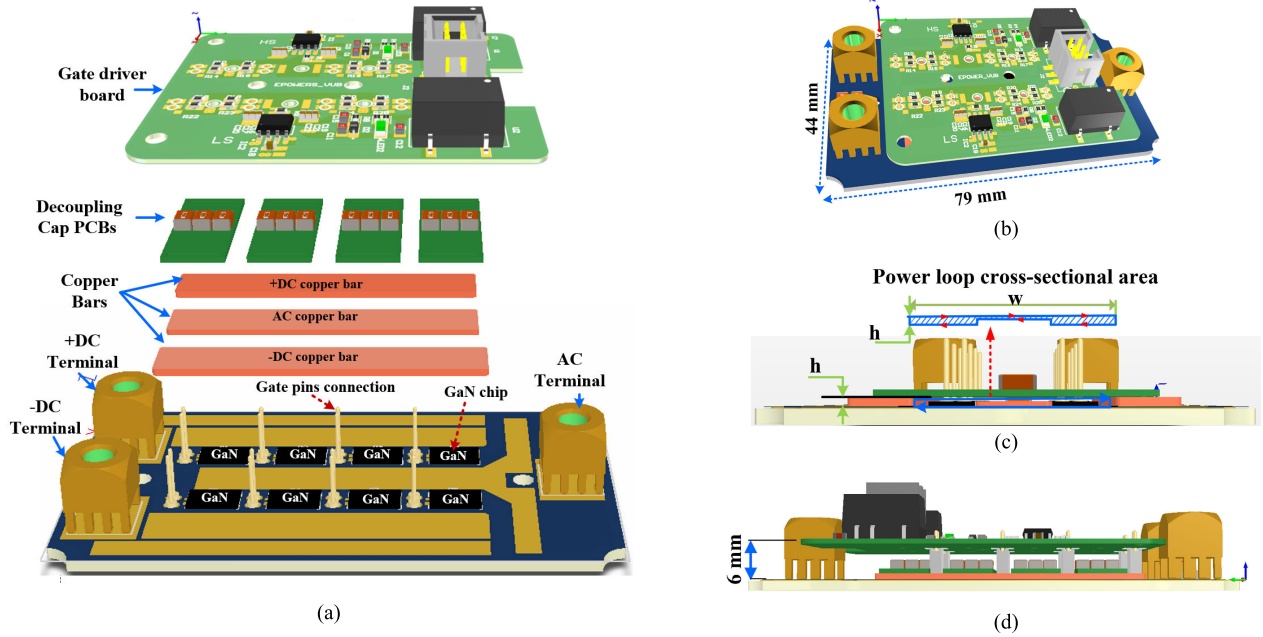


Fig. 5. Three-dimensional CAD model of the proposed GaN module. (a) Exploded view of package structure. (b) Isometric view. (c) Cross-sectional area view of power loop. (d) Side view.

development costs by minimizing the need for multiple physical prototypes [31], [32].

First, multiple PCB layout configurations for paralleling GaN devices are created using Altium Designer. The layout is then exported in ODB++ format and imported into Ansys SIwave, where adjustments to the geometry, layer thicknesses, and materials are made. Following this, Ansys Q3D is employed to calculate the stray inductances in the power and driver loops. Second, the DPT simulation is performed in LTSPICE to analyze the switching characteristics and identify potential instability issues. The simulation utilizes GaN device SPICE models provided by manufacturers, along with parasitic elements extracted from Ansys Q3D. By examining the switching behavior of V_{DS} and V_{GS} in the GaN module, the risk of power module failure due to high parasitic inductance and an imbalanced PCB layout can be effectively predicted. Third, the thermal performance of the power module is evaluated using the Ansys Icepak subsimulation tool. Finally,

considering both switching and thermal performance criteria, the optimal design solution is selected for fabrication.

B. Proposed Power Module Structure and Layout Considerations

In order to guarantee dynamic and static current sharing between parallel chips and ensure the reliability of high-power modules, several crucial factors have been considered in this design, including maintaining low and symmetrical power/gate loop inductance, achieving high thermal conductivity, and withstanding mechanical stress. Fig. 4 depicts the overall circuit diagram of the power module integrated with the gate driver, highlighting key parasitic inductances listed in Table II.

Fig. 5(a) presents a detailed exploded view of the structure, while Fig. 5(b) illustrates the compactness of the proposed half-bridge power module as demonstrated by the assembled 3-D model. The proposed design employs a novel hybrid IMS PCB structure with a vertical power loop configuration to

TABLE II
KEY PARASITIC INDUCTANCES CONSIDERED

Symbol	Name
L_D	+DC inductance of decoupling capacitor PCB
L_S	-DC inductance of decoupling capacitor PCB
L_{DC+}	+DC inductance of dc-link capacitor trace
L_{DC-}	-DC inductance of dc-link capacitor trace
L_{dd}	Drain inductance
L_{QS}	Quasi-common source inductance
L_g	Driving-gate inductance
L_{sg}	Driving-source inductance

enhance flux cancellation and heat dissipation capabilities. A total of eight GS6100P GaN chips are mounted on a single-layer IMS board to configure a half-bridge structure. Unlike previous hybrid PCB structures, where decoupling capacitors and gate driver ICs are placed together on the same FR4 PCB and connected to the power board via multipin headers or copper pins [16], [17], [29], the proposed design adopts a different approach. In this design, the decoupling capacitor boards are separated, evenly distributed, and vertically connected to each high-side and low-side GaN chip pair through +dc and -dc copper bars. This results in minimized stray inductance compared to previous structures. By using low-profile copper bars and 0.6-mm two-layer capacitor PCBs, the cross-sectional area of the commutation loop is significantly reduced, as illustrated in Fig. 5(c). This proposed design also minimizes the variation in quasi-common source inductance between each pair of high- and low-side GaN devices. The copper bars for +dc, -dc, and ac further improve the power module's capacity to handle high currents. Fig. 5(d) shows a side view of the module, where 6 mm brass head pins connect the gate and source terminals of each GaN device to the gate driver board. This ensures that the gate signals are evenly delivered to the switches while preventing any intersection of dV/dt and dI/dt power paths.

The materials and key components of the power board are listed in Table III.

C. Optimal Power Loop Inductance of Proposed Packaging Structure

Thanks to the concept of using copper bars for connecting vertically decoupling capacitor boards, the proposed structure offers low power loop inductance. However, there is still room for optimizing the geometry of the structure. Fig. 6 demonstrates the cross-sectional view of the proposed power module configuration. The Ansys Q3D FEA tool is used to quantify the commutation loop inductance of each combination. Fig. 6 shows the preliminary parameters, including fixed and adjustable parameters used for optimization. Assuming that fringing, skin, and proximity effects are negligible and adopting the classical equation for the approximated inductance of PCB [33], [34], the inductance of the power loop can be calculated as in the following equation:

$$L_{loopj} \cong \mu_r \mu_o \frac{h}{w} l = \mu_r \mu_o \frac{h_3 w_4}{w_6}, \quad \text{if } h_3 \ll w_6 \quad (1)$$

TABLE III
MATERIALS AND PARAMETERS OF THE PROPOSED POWER MODULE

IMS structure parameters	Materials	Values
Power board	IMS × 1	
○ Dimension		44×79 mm
○ Numbers of layers		Single layer
○ Conductive layer	Copper	3oz-105 μm
○ Metal base	Aluminium	2 mm
○ GaN devices (GS61008P)		100V/ 90A
Decoupling capacitor PCB	FR4-PCB × 4	
○ Dimension		20×8 mm
○ Numbers of layers		2-layers
○ Conductive layer	Copper	2 oz-70 mm
○ PCB thickness		0.6 mm
○ Capacitor	X7S	3×1 μF

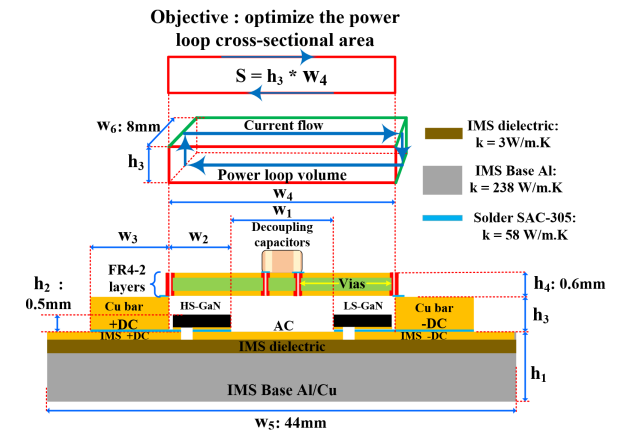


Fig. 6. Internal cross section of the power structure with the preliminary geometrical parameters.

where μ_r represents the relative permeability of the FR4 material, μ_o is the permeability of air, h_3 is the distance between the IMS copper layer and the bottom layer of the decoupling PCB, w_4 is the length, and w_6 is the width of the decoupling PCB. According to (1), the most dominant parameters influencing the power loop inductance of the proposed structure are h_3 , w_4 , and w_6 . Table IV outlines the geometrical parameters, constraints, and design guidelines necessary for maximizing the electrical and thermal performance of the proposed design. Fig. 7(a) depicts the first optimal solution with a cross-sectional area of 18×0.65 mm. The proposed design exhibits a considerably shorter height from the IMS substrate to the decoupling capacitors compared to the latest commercial GaN power modules manufactured by Infineon (formerly GaN Systems) [29], which typically range from 2 to 4 mm. The second optimal design solution is shown in Fig. 7(b). Compared to the first solution in Fig. 7(a), the length w_4 can be reduced to 15 mm by attaching an additional copper bar to the ac trace to increase current-carrying capability. The ac copper bar notably enhances the flux cancellation effect between the ac and $\pm$ dc traces, thereby further minimizing the power loop inductance.

As shown in Fig. 7(c), the third option is to use a DBC substrate instead of an IMS substrate without modifying the structure and geometrical parameters. This solution clearly

TABLE IV
PARAMETERS AFFECTING POWER MODULE PERFORMANCES

Geometrical parameters	Constrains	Design rules to improve	
		Power loop inductance	Heat transfer
Height of substrate (IMS/DBC)	$h_1 \geq 1.6$ mm Mechanical stress	N/A	- -
Height of GaN devices	h_2 Fixed at 0.5 mm (GS61008P)	-	N/A
Height of +DC/-DC bars	$h_3 > h_2$	- - -	N/A
Thickness of Cap PCBs	$h_4 \geq 0.4$ mm	-	N/A
Distance between HS and LS devices	$w_1 (\geq 200 \text{ Arms} \leq 300 \text{ Arms})$	- - -	N/A
Width of GaN devices	w_2 Fixed at 4.6 mm (GS61008P)	-	+
Width of +DC/-DC bars	w_3	N/A	N/A
Distance between +DC&-DC copperbars	$w_4 \geq w_3 + 2w_2$	- - -	N/A
Width of IMS/DBC substrate	$w_5 > w_3 + w_4$ < 50 mm	N/A	++
Width of Cap PCBs	$w_6 \geq 7.55$ mm ≤ 10 mm (7.55 mm-length of GS61008P)	+++	N/A

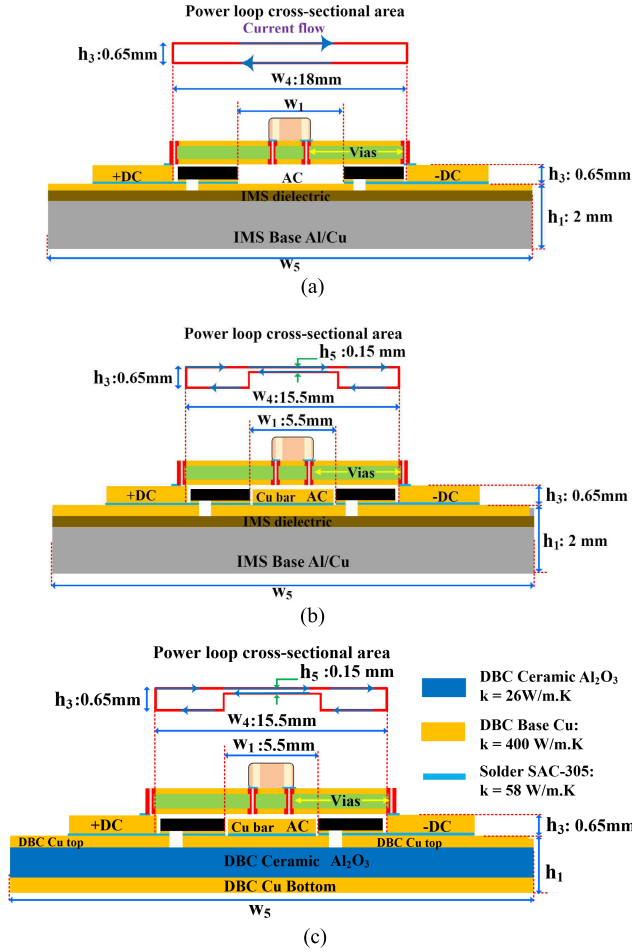


Fig. 7. Internal cross section of the proposed power structure with different geometry configurations. (a) Optimized IMS structure without ac copper bar. (b) Optimized IMS structure with ac copper bar. (c) Optimized DBC structure with ac copper bar.

demonstrated better thermal performance but comes at a higher cost.

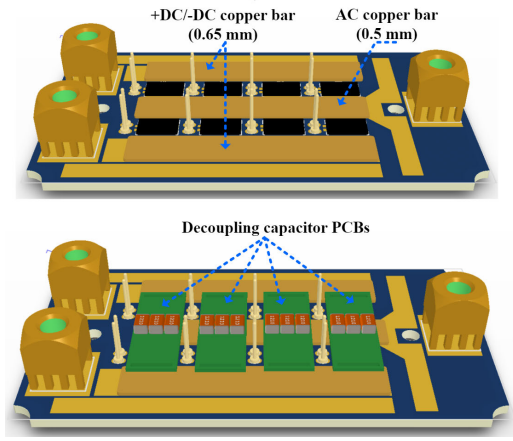


Fig. 8. Three-dimensional CAD model of the proposed IMS power board with optimized geometrical parameters and configuration.

After optimizing the geometrical configuration and parameters as depicted in Fig. 7(b), the final 3-D computer aided design (CAD) model of the proposed power module is constructed, as shown in Fig. 8.

In this design, low-profile copper bars with dimensions of $40 \times 6 \times 0.65$ mm are utilized for $\pm dc$ and $40 \times 5.5 \times 0.5$ mm for ac. As compared to existing vertical hybrid structures, the proposed structure can offer the smallest gap space between decoupling capacitors and power module substrate.

III. GATE DRIVER DESIGN CONSIDERATION FOR PARALLELING GAN DEVICES

Due to the mutual coupling between power and gate driver modules, fast transients in the power loop can significantly impact the gate-source voltage in the gate driving loop, and vice versa. Therefore, their designs must be considered simultaneously. Achieving robust gate driving signals becomes particularly challenging when driving parallel GaN devices. This section identifies the key interactions between the gate driver and power loop that increase the risk of failure in the power module. Design guidelines and the proposed gate driver layout solution are then presented.

A. Influencing Factors Linked to the Stability of the Gate Driver During Turn-On and Turn-Off Switching Transients

Fig. 9 illustrates the equivalent circuit of low-side parallel switches operating synchronously during the transient period. During the turn-on period in the saturation region, Kirchhoff's voltage law is applied to the gate loop, as illustrated in Fig. 9(a).

Then, the gate-source voltage V_{gsj} and drain-source current i_{Dj} of low-side GaN devices can be expressed as in the following equation:

$$v_{gsj} = v_{drv_ON} - i_{gj}(R_{gj} + R_{sj}) - (L_{gj} + L_{sgj}) \frac{di_{gj}}{dt} - M_{gsj} \frac{di_{Di}}{dt} - L_{sgj} \frac{d\Delta i_D}{dt} - \Delta I_D R_{sj} \quad (2)$$

where g_{fsj} is the transconductance, and M_{gsj} is the mutual inductance between quasi-common-source inductance L_{Qsj}

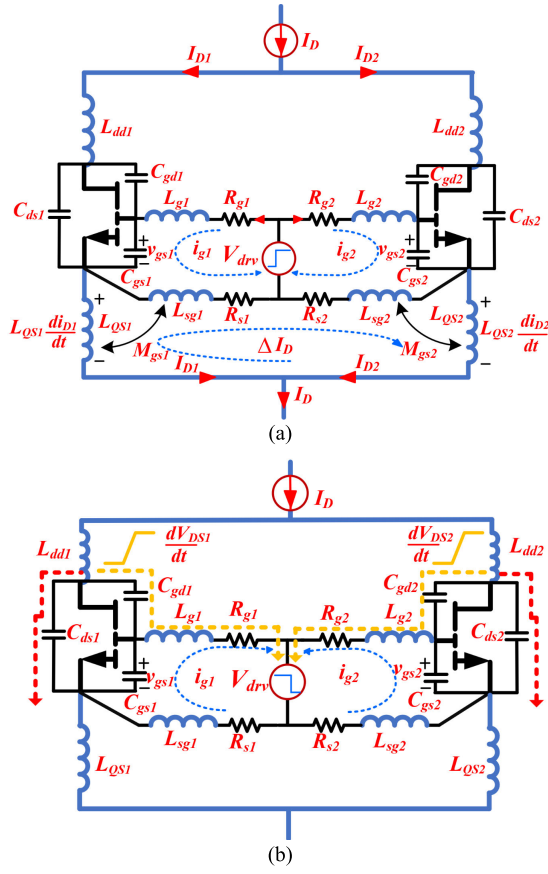


Fig. 9. Equivalent circuit of parallel devices during switching transient. (a) di/dt turn-on period. (b) dV/dt turn-off period.

and gate loop inductance

$$\begin{cases} i_{Dj} = g_{fsj}(V_{gsj} - V_{th}), & V_{gsj} > V_{th}, \quad j = 1, 2 \\ \Delta i_D = \left(L_{Qs1} \frac{di_{D1}}{dt} - L_{Qs2} \frac{di_{D2}}{dt} \right) / (Z_{s1} + Z_{s2}) \\ Z_{s1} = R_{s1} + sL_{sg1}, \quad Z_{s2} = R_{s2} + sL_{sg2}. \end{cases} \quad (3)$$

It can be deduced from (2) and (3) that the key factors causing oscillations in the gate loop voltage are the mismatch between quasi-common source inductances (L_{Qsj}), which leads to a dynamic circulating current Δi_D , and the mutual inductance (M_{gsj}) during the high di/dt switching period, as illustrated in Fig. 9(a). Therefore, to suppress the ringing gate-source voltage, it is crucial to achieve both symmetrical power loop inductance and quasi-common source inductances L_{Qsj} across parallel GaN devices. In addition, minimizing these inductance values as much as possible will help reduce the negative effects of di_D/dt [22].

During the turn-off transient, the I_D current decreases and charges the output capacitor of the low-side switch. Meanwhile, high dV/dt caused by the ultrahigh-speed switching of GaN devices results in a feedback current flowing from the drain to the source terminal of the low-side switch through the Miller capacitance C_{gd} , as depicted in Fig. 9(b). In this transient period, effects of high dV/dt predominate over di/dt . Thus, the gate-source voltage of GaN switches can be approximated as in the following equation with the assumption of

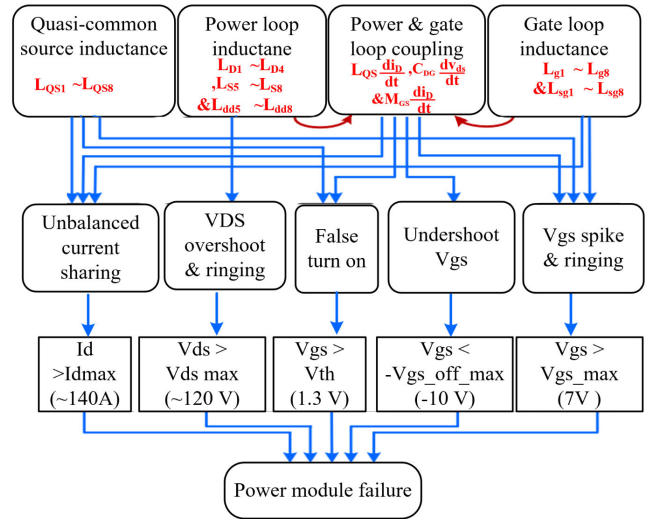


Fig. 10. Critical factors resulting in failure of paralleled GaN (GS61008P) chips during dynamic transient period.

$$M_{gsj} \approx 0 \text{ and } \Delta I_D \approx 0:$$

$$\begin{cases} v_{gsj} = v_{drvOFF} + C_{gdj} \frac{dv_{DSj}}{dt} (R_{gj} + R_{sj}) \\ \quad + (L_{gj} + L_{sgj}) C_{gdj} \frac{d^2 v_{DSj}}{dt^2} \\ i_{gj} = C_{gdj} \frac{dv_{DSj}}{dt}, \quad j = 1, 2. \end{cases} \quad (4)$$

The Miller effect results in a voltage spike of v_{gsj} . If v_{gsj} exceeds v_{th} , false triggering can occur, potentially leading to the destruction of the power module due to a short circuit incident. According to (4), this risk can be mitigated by minimizing parasitic capacitance C_{gdj} , parasitic gate loop inductance ($L_{gj} + L_{sgj}$), splitting gate resistance ($R_{gj} + R_{sj}$), or applying a negative turn-off voltage of v_{drvOFF} .

According to the analysis above, the crucial factors contributing to the failure of paralleling multiple GaN devices can be summarized as shown in Fig. 10. Unbalanced quasi-common source inductance and power-to-gate-loop coupling are identified as the most critical issues resulting in the failure of power modules. It can be indicated that the effect of high di/dt is prominent in low-voltage (i.e., 48 V) and high-current power module applications. Meanwhile, high-voltage (400 V) and high-current power modules are significantly affected by both the Miller effect and quasi-common source inductances due to the combined influence of dV/dt and di/dt .

B. Gate Driver Design Consideration

Using a separate gate driver IC for each individual GaN chip in a parallel configuration is a straightforward solution to driving multiple GaN devices. This technique simplifies circuit arrangement and allows for the adoption of gate drivers with reduced power needs. However, manufacturing tolerances and variations in IC temperatures might result in unequal propagation delays among the driver ICs. Consequently, small timing errors might cause an imbalance in the transient drain-source

current distribution. Thus, for driving multiple GaN devices, it is preferable to employ a single commercial gate driver IC with a high pulse sink/source current rating and fast transient periods [35], [36], as depicted in Fig. 4.

1) *Negative Voltage Value Selection*: Due to the quite low turn-on threshold voltage, two bipolar isolated power supplies are used to provide negative turn-off voltages for both the high-side and low-side drivers, preventing unintended turn-on caused by the Miller effect. However, a large negative gate–source voltage V_{gs_off} leads to a significant voltage drop V_{SD} during reverse conduction mode, as explained in (5). This increases reverse conduction losses during the dead-time period, as expressed in (6)

$$\begin{cases} V_{SD} = V_{th} - V_{gs_off} + I_D \times R_{DS(on)} \\ -V_{gs_off_max} \leq V_{gs_off} \leq 0 \end{cases} \quad (5)$$

$$P_{reverse_loss} = V_{SD} \times I_D \times t_{dead} \times f_{sw} \times 2. \quad (6)$$

Furthermore, a high value of V_{gs_off} increases the risk of damaging GaN during transient conditions, as it may surpass the maximum allowable negative voltage $V_{gs_off_max}$ of -10 V specified for the GS61008P.

2) *Gate Resistors Selection*: Equation (1) is rearranged and simplified into the following equation by assuming $M_{gsj} \approx 0$ and $\Delta I_D \approx 0$:

$$\begin{cases} V_{drv} = V_{gsj} + i_{gj} R_{gTj} + L_{gTj} \times \frac{di_{gj}}{dt} \\ i_{gj} = C_{gsj} \times \frac{dV_{gsj}}{dt}, \quad j = 1, 2, 3, 4 \\ R_{gTj} = R_{gj} + R_{sj} \\ L_{gTj} = L_{gj} + L_{sgj}. \end{cases} \quad (7)$$

The transfer function, representing the relationship between V_{gsj} and V_{drv} , is obtained in the following equation by taking the Laplace transform of (7):

$$G(s) = \frac{V_{gsj}(s)}{V_{drv}(s)} = \frac{\frac{1}{L_{gTj}C_{gsj}}}{s^2 + \frac{R_{gTj}}{L_{gTj}}s + \frac{1}{L_{gTj}C_{gsj}}} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}. \quad (8)$$

The turn-on gate resistance is determined according to the stray gate loop inductance. To suppress ringing with a proper switching speed, the oscillation must be damped quickly. Based on the standard second-order ordinary differential equation (ODE), the critical damping ratio ζ of 0.707 is selected. Thus, the required total resistance value must be satisfied by the following equation:

$$R_{gTj_ON} \geq 2\zeta \sqrt{\frac{L_{gTj}}{C_{iss}}}. \quad (9)$$

As previously indicated, to prevent the bouncing of the gate–source voltage V_{gsj} due to high dV/dt during the turn-off transient, the turn-off gate resistance is selected to ensure that the gate–source voltage does not exceed the gate-threshold voltage V_{th} of 1.3 V. The turn-off gate resistance R_{off} is the sum of R_g and R_s , calculated using (10) [23]. In 48-V motor

drive applications employing 100-V GaN devices, a maximum dV/dt is selected as 10 V/ns, and reverse transfer capacitance ($C_{gd} = C_{rss}$) is 12 pF, according to the datasheet of GS61008P

$$R_{off} \leq \frac{V_{th}}{C_{gd} \times \frac{dV_{ds}}{dt}} = \frac{1.3 \text{ V}}{12 \text{ pF} \times 10 \frac{\text{V}}{\text{ns}}} = 10.8 \Omega \quad (10)$$

where R_{off} is the total resistance of the splitting resistors of R_g and R_s . Hence, the split turn-on resistor of 4.7Ω and turn-off resistor of 4Ω are utilized to suppress the circulating current in the parallel gate loop and adjust V_{DS} voltage spike.

C. Design Rules Applied in the Proposed Gate Driver Design

As aforementioned in Section III-A, the impact factors on the stability of the gate driver are identified. This section provides the design rules and solutions to address the challenges of the gate driver module as follows.

- 1) It is advisable to use split driving outputs to independently manage slew rates during on/off transients. In addition, reducing resistance in the turn-off loop helps mitigate the Miller effect, and applying a negative turn-off voltage of -3 V extends the safe margin for the threshold voltage.
- 2) To ensure mutual decoupling between the gate and power loops, it is crucial to avoid overlapping between the switching node path and the gate-driving path. In addition, common-source inductance can be minimized by using GaN devices equipped with Kelvin pins. To further reduce crosstalk caused by high dV/dt and di/dt through parasitic capacitances and mutual inductances, it is recommended to separate the gate driver board from the power board.
- 3) Maximize the overlap of the PCB layout between the turn-on path and the return path while minimizing the distance between two layers to enhance effective flux cancellation for reducing gate loop inductance.
- 4) Achieving uniform gate-driving loop inductances is possible by using a symmetrical PCB layout for the gate driver board. Furthermore, separate resistors for turn-on and turn-off ($R_{g1}-R_{g8}$, $R_{s1}-R_{s8}$) are positioned at both the gate and source terminals of each GaN transistor, minimizing the impact of circulating currents within the gate driver loops.

Fig. 11 illustrates the circuit diagram and PCB layout of the gate driver designed for the proposed half-bridge power module. The turn-on gate current path is depicted in Fig. 11(a) and (b). The total gate resistance is the sum of the common ON-resistance R_{on} , splitting resistance of R_g , and R_s . Likewise, the turn-off gate current path is depicted in Fig. 11(c) and (d), with a total turn-off resistance comprising R_g and R_s . As shown in Fig. 11(b) and (d), it can be observed that the current in both turn-on and turn-off paths flows in opposite directions in the overlapping top and bottom layers, resulting in a reduction of gate loop inductance. Furthermore, to mitigate the inductance mismatch, the lengths of the PCB layout traces from the gate driver IC to each gate–source connection of GaN devices should be approximately identical.

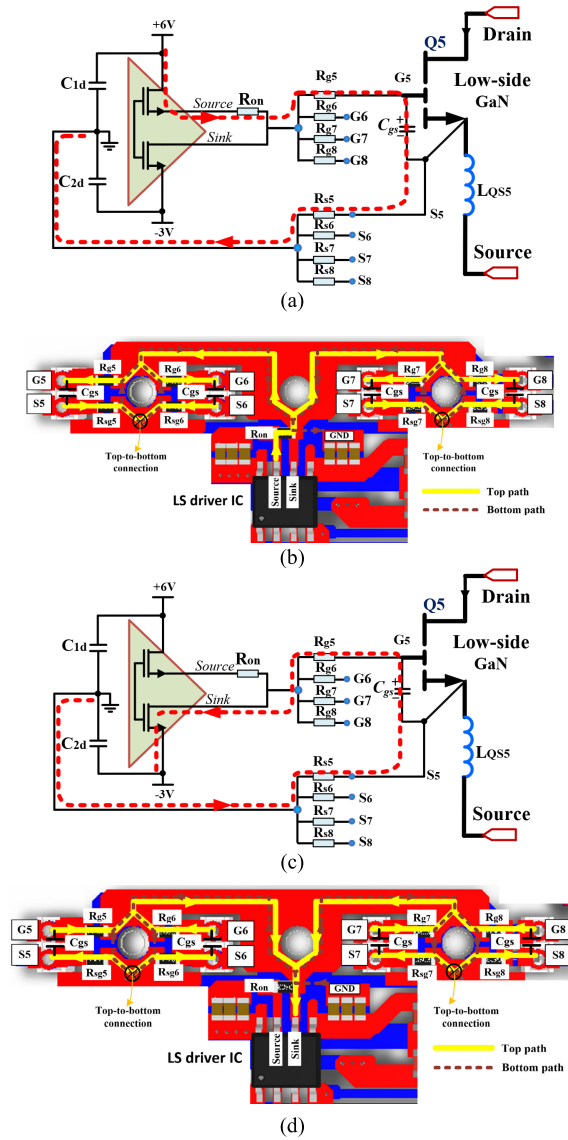


Fig. 11. Circuit diagram and PCB layout of the low-side gate driver board. (a) Turn-on driving path. (b) Turn-on PCB layout. (c) Turn-off driving path. (d) Turn-off PCB layout.

Table V includes the main components with their corresponding values.

The isolated power supplies provide $+6/-3$ V for gate driver ICs Si8271, which have a split sink and source output driving terminal. Thus, R_{on} and R_{off} can control the turn-on and turn-off slew rates separately. Since the gate driver PCB is connected to the IMS substrate through Kelvin gate-source brass pins and stacked on top of the power board, the gate driver PCB is physically separated from the power IMS board. This arrangement minimizes both inductive and capacitive coupling. In applications requiring higher switching speed, the gate driver IC 1EDB7275F can be an alternative solution, with detailed parameters provided in Section A of the Appendix.

IV. PARASITIC EXTRACTION AND THERMAL SIMULATION

A. Parasitic Inductance Extraction and Simulation of DPT

In this section, Ansys Q3D is employed to extract the power and gate loop inductances of the proposed power module

TABLE V
GATE DRIVER COMPONENTS SPECIFICATIONS

Gate driver parameter	Values
Gate driver IC	Si8271GB
○ Input voltage	5V
○ Source/sink current (typical)	1.8 A/4 A
○ Split output	Sink/Source
Bipolar Power Supply	
○ Power supply module	Xpower
○ Power	2 W
○ In/Output voltage	+5V/+9V
○ Drive voltage	+6.2V/-3V
Turn-On/Off resistors	
○ Common on-resistor	10 Ω
○ Split on-resistor	4.7 Ω
○ Split off-resistor	4 Ω

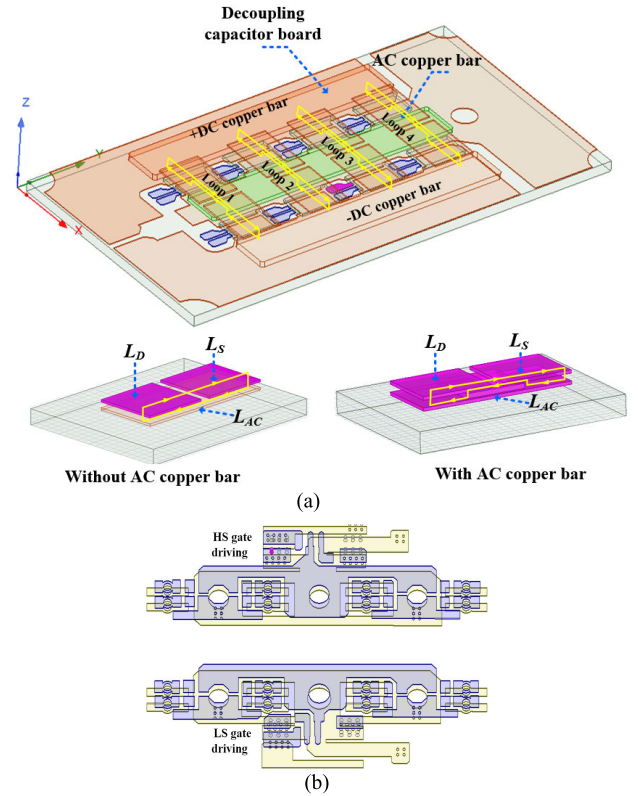


Fig. 12. Q3D models for parasitic inductances extraction. (a) Power module model and simplified power loop models. (b) Gate driver board.

following the framework outlined in Section II-A. As shown in Fig. 12, the 3-D models, including the copper layout of the power and gate driver modules, are imported into the Ansys SIwave/Q3D simulation environment. Once the material and geometric dimensions are established, the current flow in the power loop is determined through the configuration of the sink and source nets. Fig. 12(a) depicts the Q3D model of the power board and simplified power loop models. The +dc and -dc inductances of the decoupling PCB are denoted by L_D and L_S , respectively, whereas L_{ac} represents the inductance of the switching node.

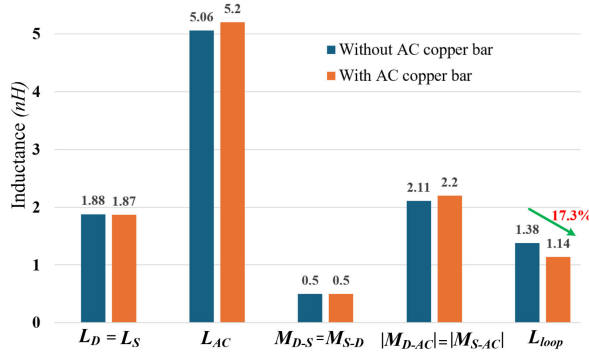


Fig. 13. Comparison of power loop inductances without and with the ac copper bars configuration.

TABLE VI
PARASITIC INDUCTANCES EXTRACTION BY Q3D

Parameter	Symbol	Values
See Table II	$L_D = L_S$	1.87 nH
AC inductance	L_{AC}	5.2 nH
D-S mutual inductance	$M_{D-S} = M_{S-D}$	0.5 nH
D-AC mutual inductance	$M_{D-AC} = M_{S-AC}$	2.2 nH
Power loop inductance	$L_{Loop1} - L_{Loop4}$	~ 1.14 nH
Gate loop inductance	L_{g_total}	~ 7 nH

The inductance value matrix can be generated from Q3D at 100-MHz solution frequency as follows:

$$\begin{bmatrix} L_D & M_{D-S} & -M_{D-ac} \\ M_{S-D} & L_S & -M_{S-ac} \\ -M_{ac-D} & -M_{ac-S} & L_{ac} \end{bmatrix}.$$

The mutual inductances M_{D-S} and M_{S-D} represent the magnetic coupling between L_D and L_S . Similarly, M_{ac-D} , M_{ac-S} , M_{D-ac} , and M_{S-ac} represent the mutual inductance of L_{ac} with L_D and L_S , respectively. The opposing currents in the power loop lead to flux cancellation, resulting in negative values for these parameters. Consequently, the power commutation loop inductance is expressed in the following equation:

$$\begin{aligned} L_{loop} &= L_D + L_S + L_{ac} + 2(-M_{D-ac} - M_{S-ac} + M_{D-S}) \\ &= 1.87 + 1.87 + 5.20 + 2 \times (-2.20 - 2.20 + 0.5) \\ &= 1.14 \text{ (nH)}. \end{aligned} \quad (11)$$

A similar inductance extraction method is applied to the gate driver Q3D model, as shown in Fig. 12(b). Subsequently, the detailed parasitic inductance values are obtained and listed in Table VI.

As shown in Fig. 13, the simulation results compare single-power loop inductance values with and without ac copper. The inductance of L_{ac} increased from 5.06 to 5.20 nH. Fortunately, (11) highlights that the most dominant factors in reducing loop inductance are M_{D-ac} and M_{S-ac} . Thus, a slight increase in these values from 2.11 to 2.2 nH will considerably enhance flux cancellation effects, leading to a reduction in the power loop inductance from 1.38 to 1.14 nH. This value is close to 1.175 nH obtained in the experimental power loop extraction in Section B of the Appendix. Most notably, a thin copper bar is placed close to the GaN chip's source terminal, nearly contacting it. Thus, the quasi-common

source inductance value can be considered the internal source inductance of the GaN chip, which has been simulated to have an ultralow common source inductance of 0.05 nH (GS66516B), as reported in [30]. To gain insight into the switching dynamics of the proposed design, a simulation of the DPT was performed using LTSPICE software. The simulation model was established with all parasitic components obtained from Ansys Q3D and the GS61008P eGaN HEMT SPICE model provided by GaN Systems. In this high-current GaN-based inverter design, it is necessary to verify the overshoot voltage of the power module at a rated voltage and current of around 48 V/350 A and an operating switching frequency f_{sw} of 20 kHz. Therefore, the load inductor for the DPT is sized as described in the following equation:

$$\begin{cases} L_{DPT} \leq (\tau_1 + \tau_2) \times \frac{V_{dc}}{I_L} \\ \tau_1 + \tau_{break} = 1/f_{sw} \end{cases} \quad (12)$$

where τ_1 , τ_2 , and τ_{break} are the first, second pulse, and pulse break durations, respectively. The pulse length can be chosen such that 70% of the desired load current is reached. During the first pulse duration, the dc-link capacitor bank stabilizes the dc bus voltage with a suitable voltage ripple ΔV_{dc} . By transferring electrical energy E_C from the capacitor bank into the magnetic energy E_L of the load inductor, the energy balance formula can be established for sizing the dc-link capacitor, as expressed in the following equation:

$$\begin{cases} \Delta E_C \geq E_L \\ = \frac{1}{2} C_{dc} V_{dc}^2 - \frac{1}{2} C_{dc} (V_{dc} - \Delta V_{dc})^2 \geq \frac{1}{2} L_{DPT} I_L^2 \\ C_{dc} \geq \frac{L_{DPT} I_L^2}{2 V_{dc} \Delta V_{dc} - \Delta V_{dc}^2} \end{cases} \quad (13)$$

where ΔE_C is the energy drop of the dc-link capacitor during the first pulse period. Typical values of ΔV_{dc} range from 1% to 5% of V_{dc} . According to (12) and (13), the load inductor and capacitor bank values are selected as 5.5 μ H and 400 μ F, respectively. Fig. 14(a) shows the DPT simulation waveforms. The first and second turn-off switching currents are 275 and 325 A, corresponding to the drain-source voltage spikes V_{DS} of 64 and 65.8 V, respectively. Thus, the overshoot voltage across the device has a sufficient margin for safe operation. In practice, this voltage spike can be controlled by adjusting the rating transient of dV/dt through the turn-off gate resistor. Fig. 14(b) shows the drain-source currents during switching transient periods. As mentioned in Section I, in high-power traction inverter applications, high dV/dt causes high voltage stress on the insulation of motor windings and bearing current issues. Typically, a higher dV/dt results in lower switching losses but necessitates a heavier and larger dV/dt filter between the inverter and motor terminals [37], [38]. Therefore, while Si/insulated gate bipolar transistors (IGBTs)-based inverters are usually designed to operate within the 5–10 kHz range, GaN HEMT devices allow an increase in switching frequency to the 10–40 kHz range. Many studies have commonly chosen a 20-kHz switching frequency for high-power

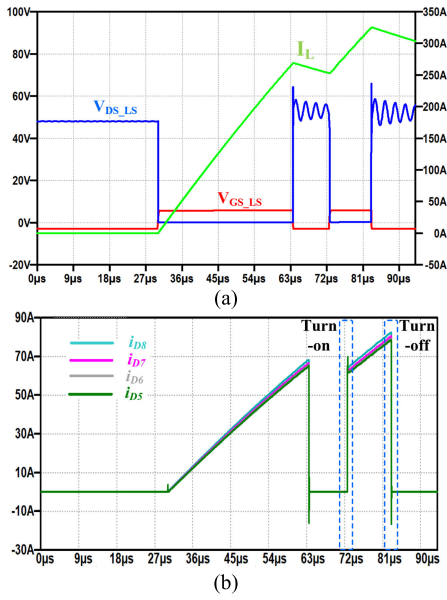


Fig. 14. LTSPICE DPT simulation result at 48 V/325 A. (a) Drain-source voltage V_{ds} , load current I_L and gate-source voltage V_{gs} . (b) Drain current I_D waveforms.

TABLE VII
THERMAL PARAMETERS OF IMS POWER MODULE

Material	Thickness	Thermal Conductivity [W/mK]
Copper	105 μ m	400
Polymeric Film	76 μ m	3
Aluminium substrate	2 mm	237.5

GaN traction applications [20], [39], [40], [41], [42]. In addition, a dV/dt value ranging from 3 to 6 V/ns is typically recommended in these cases [43].

The dynamic current mismatch in asymmetrical paralleled GaN packaging [15], [21], [44] is chosen as the baseline reference for comparison with the symmetrical power loop configuration in the simulation results, presented in Fig. 15.

In Fig. 15(a) and (b), the current imbalance varies depending on the different power loop inductances resulting from the asymmetrical layout. The GaN device exhibits the smallest power loop inductance, enduring the highest current stress, while others with larger loops experience lower levels of current stress. In the symmetrical structure approach, identical power loop inductances result in uniform dynamic currents during turn-on and turn-off switching transient periods, as illustrated in Fig. 15(c) and (d).

B. Thermal Performance Simulation

The thermal performance of the proposed GaN-based power module is investigated by FEA software. In order to investigate the thermal characteristics of the power module, the electrical CAD (ECAD) model is imported from Altium Design (or Ansys SIwave) to Ansys Icepak software. The thermal properties of the power module components are set according to the values shown in Table VII. The power loss in each GaN device is set at 15 W, representing a per-phase loss of 120 W. The ambient temperature is set to 40 $^{\circ}$ C, and the heat transfer coefficient of the power module is set at 1000 W/m 2 $\cdot$ K,

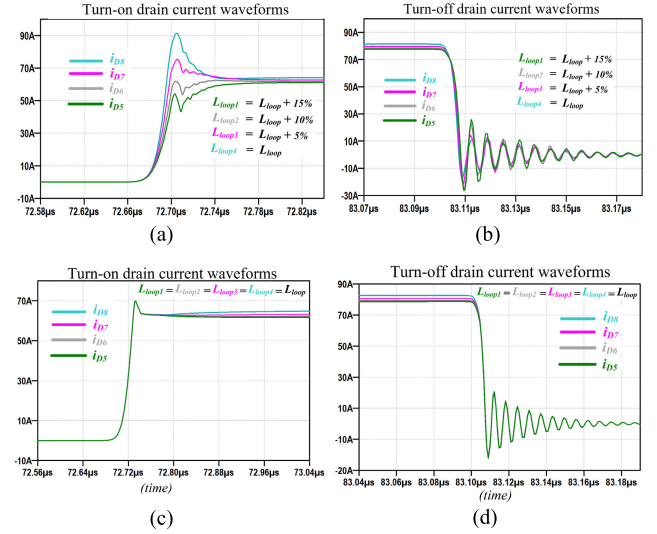


Fig. 15. LTSPICE simulation results of dynamic currents in low-side four paralleled GaN devices during turn-on and turn-off periods. (a) and (b) Asymmetrical power loop configuration. (c) and (d) Symmetrical power loop configuration.

which is equivalent to the cooling provided using an air-cooled heat sink. The steady-state analysis indicates that the thermal performance of the GaN devices settles within the safe operating range for GaN devices, as shown in Fig. 16. The proposed GaN-based hybrid packaging structure has shown thermal performance comparable to that reported in [23].

Fig. 17 illustrates the maximum junction temperature of the GaN devices under varying power losses and heat transfer coefficients. The results indicate that both air-cooled and liquid thermal management systems are adequate for maintaining acceptable power loss levels and providing a sufficient temperature margin for safe operation.

V. POWER MODULE FABRICATION PROCESS

As mentioned in Section II-C, despite the DBC-based solution offering higher thermal conductivity, the IMS-based design is chosen for laboratory prototyping to validate the proposed concept. The fabrication procedure of the power module is illustrated in Fig. 18 and can be summarized as follows.

Step 1: The fabrication process begins by applying SAC305 solder paste, composed of 96.5% tin, 3% silver, and 0.5% copper, to a 3-oz IMS substrate through a stencil that is 100 μ m thick. Afterward, eight GaN chips are mounted on the board, with 16 brass head pins connecting the gate-source terminals of each chip. Following this, two $\pm$ dc copper bars (40 $\times$ 6.0 $\times$ 0.65 mm), an ac copper bar (40 $\times$ 5.5 $\times$ 0.5 mm), and power terminals are placed on the IMS board. The IMS module is then subjected to the reflow soldering process within 10 min, which heats it to a recommended maximum temperature of 220 $^{\circ}$ C [45]. Finally, a 0.08-mm-thick layer of Kapton tape is applied to cover the ac copper bar, enhancing electrical isolation.

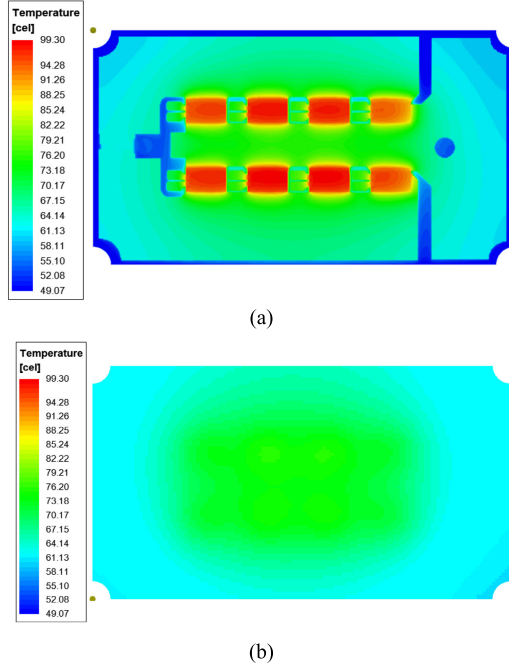


Fig. 16. Simulated thermal distribution of proposed power module at 15-W power loss per individual device, $T_{amb} = 40^\circ\text{C}$ and heat transfer coefficient of $1000\text{ W/m}^2 \cdot \text{K}$. (a) Top view. (b) Bottom view of power module.

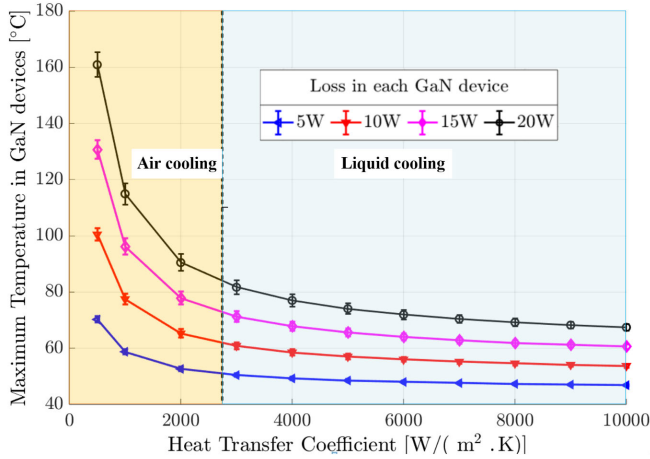


Fig. 17. Junction temperature of GaN devices corresponding to variation of heat transfer coefficients and power losses.

Step 2: Decoupling capacitor PCBs with a copper thickness of 1 oz and a total thickness of 0.6 mm are presoldered with three $1\text{ }\mu\text{F}/100\text{-V}$ decoupling capacitors on the top side. Each capacitor PCB is then placed onto the +dc and -dc copper bars, ensuring that every pair of high-side and low-side GaN switches is equally distributed by its own decoupling capacitor PCB. The PCBs are subsequently soldered onto the main power module using a reflow process with Sn63/Pb37 solder paste, which melts at 183°C . This step completes the assembly of the power IMS board.

Step 3: All surface-mount device (SMD) components, including gate driver ICs, isolated power supplies, and passive components, are soldered onto the

TABLE VIII
COMPONENTS AND EQUIPMENT FOR EXPERIMENTAL SETUPS

DPT, Buck & Inverter	Descriptions	Values
○ Capacitor bank	Film capacitor	500 μF
○ DPT inductor, L_{DPT}	Air core inductor	$\sim 5.5\text{ }\mu\text{H}$
○ Buck inductor, L_{BUCK}	Air core inductor	76 μH
○ Output capacitor, C_O	Film capacitor	200 μF
○ R/L load for inverter		$0.15\Omega/125\mu\text{H}$
○ DSP control board	TMS320F28379D	N/A
Equipment	Parameters	
○ DC power supply		80 V-500 A
○ Electronic load	Bidirectional	500 V-150 A
○ Oscilloscope	TDS5054B	500 MHz
○ Differential probe	Tektronix P5205A	100 MHz
○ Passive probe	TPP0101	100 MHz
○ Current probe	TCP0150A	50 MHz-500 A _{peak}

gate driver PCB, which has a thickness of 1 mm. Sn63/Pb37 solder paste and a maximum temperature setting of 183°C are used for the reflow soldering process.

Step 4: Finally, the gate driver board is attached to the IMS power board via gate-source nail-head pins. Fig. 18 shows the fully assembled power module, which has a low profile and high power density, with overall dimensions of $79 \times 44 \times 10\text{ mm}$. The total height of the power module can be further reduced by selecting lower-profile connection terminals.

Using simple -dc/+dc/ac copper bars reduces the copper layer thickness of the IMS board, allowing for the use of standard single-layer IMS, which is much less expensive than multilayer FR4 PCBs or alternatives with thicker copper layers. In addition, placing most components on one side of the power substrate, along with the decoupling capacitor PCBs and the gate driver board, simplifies the reflow soldering process. Furthermore, the wire-bondless packaging and straightforward fabrication process make this proposed design both cost-effective and dependable for paralleling multiple GaN chips. This laboratory prototype can be implemented without the need for specialized equipment or advanced techniques.

VI. EXPERIMENTAL VALIDATION

In this section, several experimental setups have been implemented to validate the performance of the proposed power module, including dynamic and static switching performances, thermal performance, and expandable power capability. The experimental setups are depicted in Figs. 19 and 20. The key components and parameters of the test setups are tabulated in Table VIII.

A. Dynamic Electrical Testing

DPT evaluations have been performed to assess the dynamic characteristics of the proposed design. Fig. 19(a) depicts the circuit diagram of the DPT testing setup for a single power module. In addition, to validate the scalability of this power module for higher power inverter applications, the

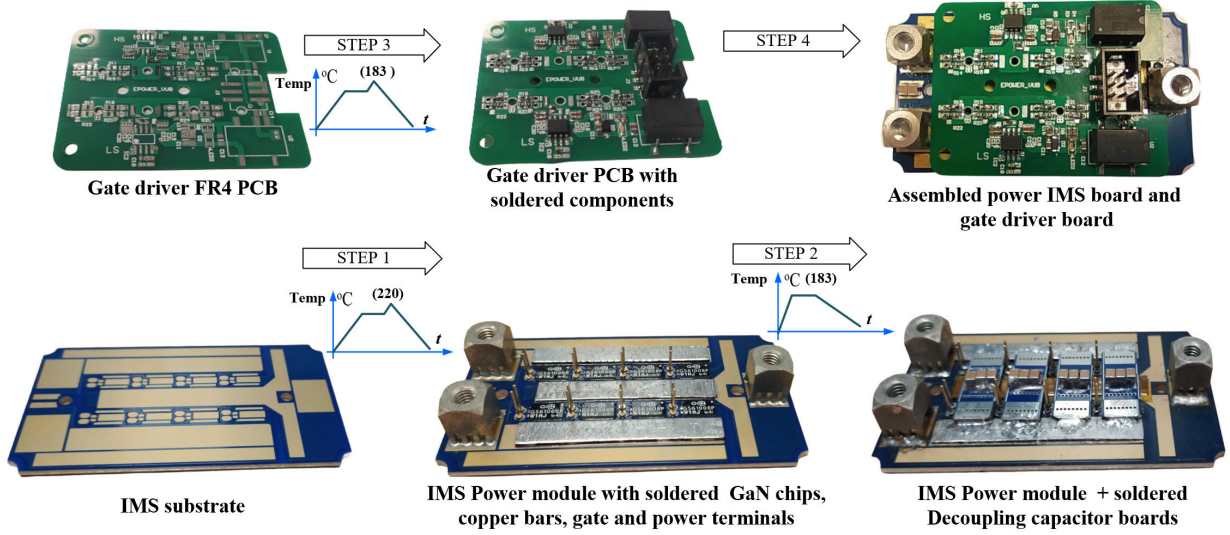


Fig. 18. Fabrication process of the proposed GaN-based power module.

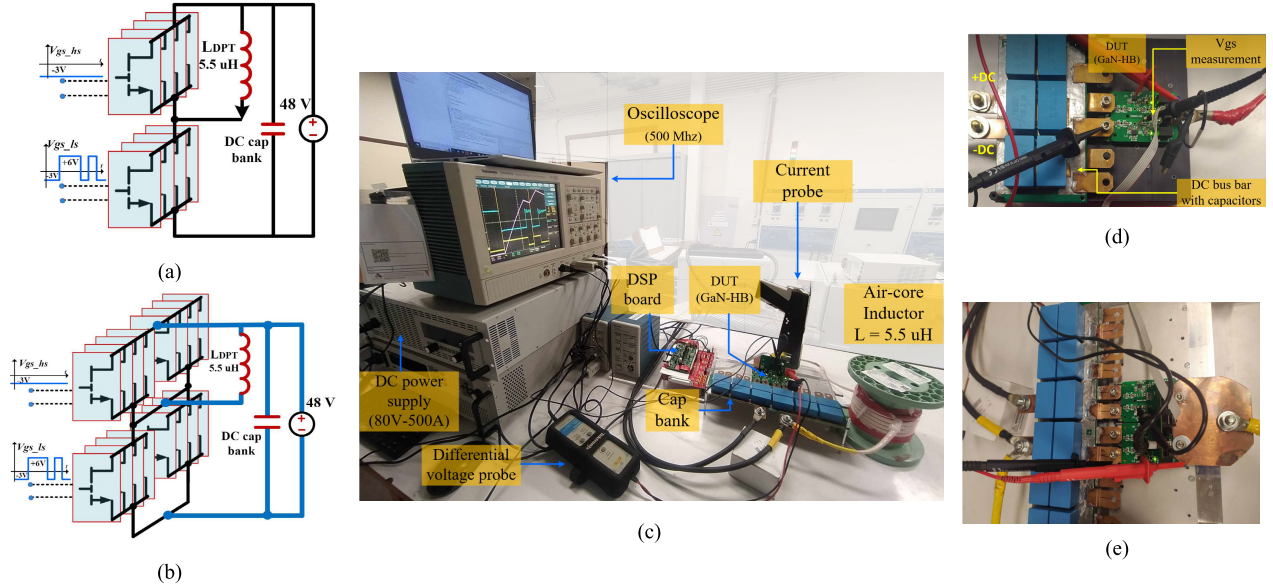


Fig. 19. Experimental DPT setups. (a) Circuit diagram for single module. (b) Circuit diagram for dual module. (c) DPT workbench. (d) Zoomed in the connection for single module. (e) Zoomed in the connection for dual module.

circuit configuration involving dual paralleled GaN modules is evaluated, as depicted in Fig. 19(b). The test platform setup is shown in Fig. 19(c), with detailed close-ups of the DPT setups for single-module and dual-module configurations provided in Fig. 19(d) and (e), respectively. To prevent magnetic saturation under high switching currents, an air-core inductor with a value of $5.5 \mu H$, as calculated in (11), is utilized as the inductive load. In addition, a $500\text{-}\mu F$ dc-link capacitor bank is used to maintain a constant voltage. The high-side devices function as freewheeling switches by continuously supplying -3 V to the gate-source pins, while the low-side GaN transistors operate as the main switches. A digital signal processing (DSP) board is used to generate the double pulse signal for the low-side gate drive IC.

The waveforms of the drain-source voltage V_{DS} at the switching node, the gate-source voltage V_{GS} , and the inductor

current I_L are captured using a differential probe, a passive probe, and a current clamp probe, respectively. These waveforms are displayed on the TDS5054B oscilloscope, as shown in Fig. 21. Table VIII contains detailed specifications of the measurement equipment. As shown in Fig. 21(a) and (b), the proposed power module has fulfilled the test successfully under hard switching conditions of $48\text{ V}/328\text{ A}$ with a single module and $48\text{ V}/400\text{ A}$ with two paralleled modules. Zooming into the detailed turn-on and turn-off transient waveforms of the single module DPT test in Fig. 21(c) and (e), the voltage spike reaches 60.8 V with a dV/dt of 3.2 V/ns .

In a dual module test, the spike voltage V_{DS} is only 49.6 V , maintaining the same dV/dt of 3.2 V/ns , as illustrated in Fig. 21(e) and (f). This test also indicates that the single module has a 3.3% overshoot voltage under 200-A switching current. In addition, the symmetrical structure results in a

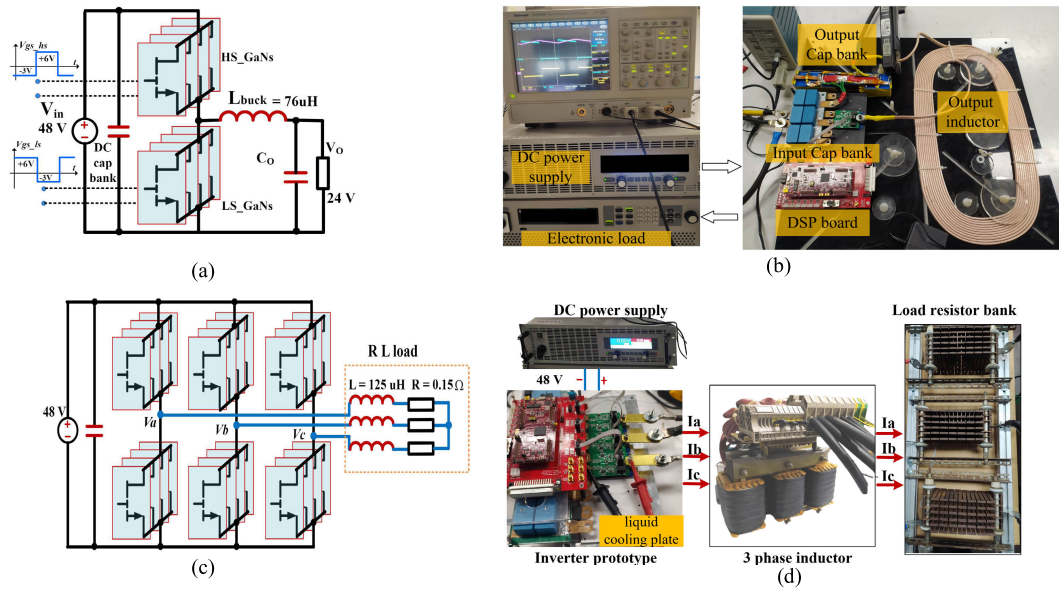


Fig. 20. Experimental setups for continuous power tests. (a) Circuit diagram of buck converter. (b) Hardware setup of buck converter. (c) Circuit diagram of open-loop three phase inverter. (d) Hardware setup of three phase inverter system.

minimal V_{GS} mismatch between the two modules. The V_{GS} waveforms are stable, with low oscillation and bouncing voltages of less than 1.2 V at both 328- and 200-A switching currents. The dynamic testing results validate the robustness of switching waveforms under high-current switching conditions. The detailed comparison of switching performance between the proposed GaN power module and the existing 48-V Si-based power modules is listed in Table X of Section C in the Appendix.

B. Static Continuous Power Testing

The aim of this test is to verify the operational performance of the proposed GaN power module under continuous power conditions. The buck-converter and three-phase inverter experimental circuit diagrams are shown in Fig. 20(a) and (c), with the parameters listed in Table VIII. Fig. 20(b) and (d) shows the prototype in the testing setups.

The power module is attached to an aluminum heat sink with dimensions of $100 \times 150 \times 25\text{ mm}$, where heat is dissipated via forced air cooling. In the 48–24 V/3-kW open-loop buck converter test, the duty cycle is set at 50% with a 200-ns dead-time duration. The inductor current I_L , drain–source voltage V_{DS} , and gate–source voltage V_{GS} waveforms of the buck converter are shown in Fig. 22 under the testing conditions of the 20-kHz switching frequency and 3-kW input power. The tested waveforms show that robust V_{DS} and V_{GS} can be achieved at an average hard switching current of 120 A. According to the DPT results in Fig. 21(b), which show a low-voltage spike of 49.6 V at a 200-A switching current, it is suggested that the proposed power module can operate safely at up to 5 kW as a buck converter with a proper cooling system.

Because of the highly compact package integration, the temperature distribution of each GaN chip is impossible to observe with the thermal camera. The thermal performance

is indirectly evaluated by observing the thermal aluminum substrate and heat sink, as shown in Fig. 23.

Fig. 24 depicts the efficiency measurement of the buck converter module under various output load currents. The maximum efficiency can reach 96.1% at a load current of 100 A. This efficiency is expected to be further improved through proper design of the coil inductor, capacitors, and selection of a suitable gate driver IC with optimal switching speed and dead-time value. However, achieving optimal efficiency for this testing is beyond the scope of this article. After confirming the operation of the proposed half-bridge module with the DPT and buck-converter tests, a three-phase inverter test setup is built according to the circuit schematic shown in Fig. 20(c). This test further investigates the potential of the power module for low-voltage and high-current applications with a liquid cooling method. As shown in Fig. 20(d), three power modules are mounted on the cold plate and supplied by a 500 A to 80-V dc power supply. The testing power is limited by the rated three-phase inductor current of 120 A.

Fig. 25 shows the phase currents and drain–source waveforms of the open loop three-phase inverter system. The phase current reaches the peak value of 160 A with an overshoot drain–source voltage of 50 V.

For heat dissipation, the chiller cooling system is connected to the inlet and outlet of the cold plate, and the flow rate is set at 10 L/min. At 5-kW power and 10 min operating duration, the thermal image of three power modules is captured, as shown in Fig. 26. The maximum temperature of the IMS aluminum substrate is only around 34 °C with the liquid cooling solution. Compared to the forced air cooling method with a heat sink, the liquid cold plate dissipates heat more effectively, as expected. Based on these test results, both the switching and thermal performances demonstrate that the proposed power module is well-suited for high-power traction inverter applications.

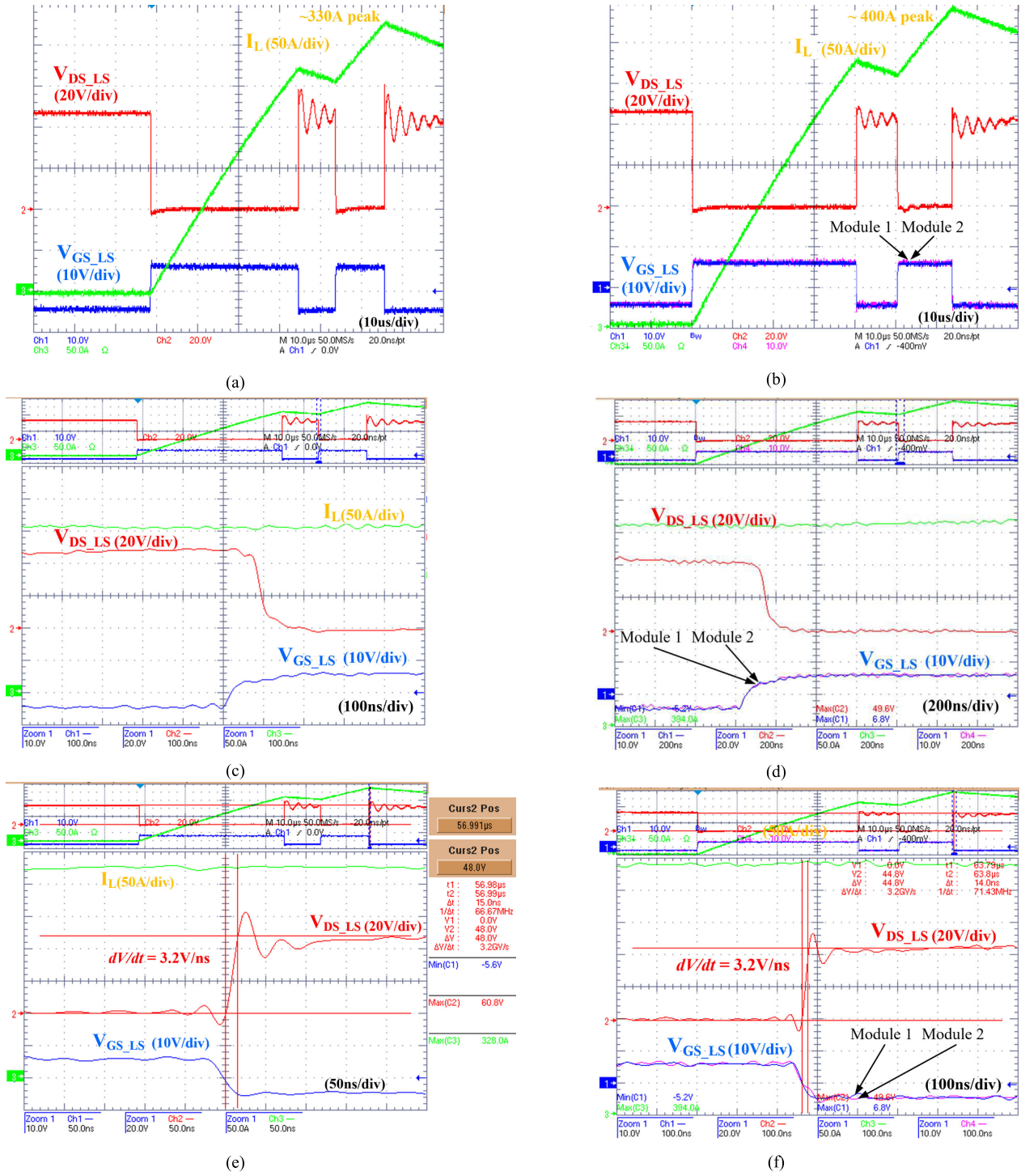


Fig. 21. Experimental DPT waveforms. (a) DPT waveforms of the single half-bridge power module at 48 V/328 A. (b) DPT waveforms of the dual half-bridge power module at 48 V/400 A. (c) Zoomed the turn-on transient in single module setup. (d) Zoomed in turn-on transient in the dual module setup. (e) Zoomed in turn-off transient in single module setup. (f) Zoomed in turn-off transient in dual module setup.

VII. DISCUSSION AND FUTURE PERSPECTIVES

The adoption of GaN power modules in future vehicle electrification brings high demands and challenging requirements in terms of power density, power rating, electrical, thermal, and mechanical stress, efficiency, and reliability. However, existing packaging structures for Si or SiC dies may not be directly applicable to GaN modules. In addition, there is a lack of research on paralleling more than two GaN devices to

address all the design challenges, particularly for high-power traction inverters. Compared to previously available methods for paralleling four GaN chips, the proposed solution meets multiple design objectives, including robust switching characteristics, high thermal conductivity, low mechanical stress, enhanced flexibility and scalability, system compactness, and cost-effectiveness, as illustrated in the radar chart in Fig. 27.

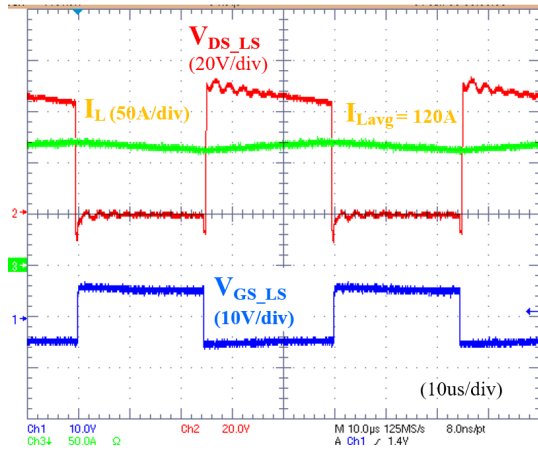


Fig. 22. Experimental waveforms of buck converter at 20 kHz to 3 kW testing condition.

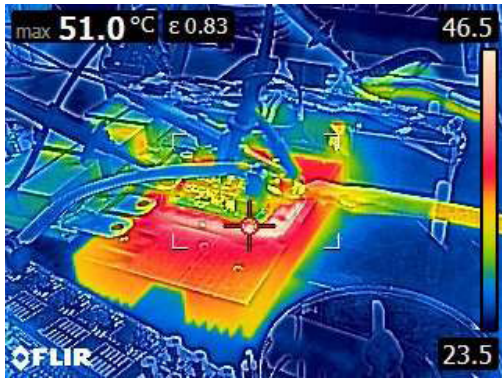


Fig. 23. Thermal image of the power module as tested in buck converter at 20 kHz to 3 kW condition.

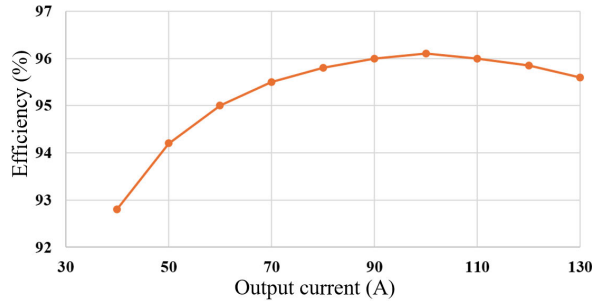


Fig. 24. Efficiency of buck converter tested at various output load currents.

In Section D of the Appendix, a mechanical simulation model of the proposed power module, including detailed material properties, is presented along with corresponding simulation results. Section E of the Appendix provides an overview of the scalability and flexibility of the inverter prototype utilizing the proposed GaN module. Although various simulations and experiments have been conducted in this study, certain limitations remain due to the scope of the research and the constraints of the measurement equipment available in the current laboratory conditions. In the future, it might be useful to implement more elaborate simulations, experiments, and hardware prototypes to validate the potential of the proposed design for high-power GaN-based inverter systems.

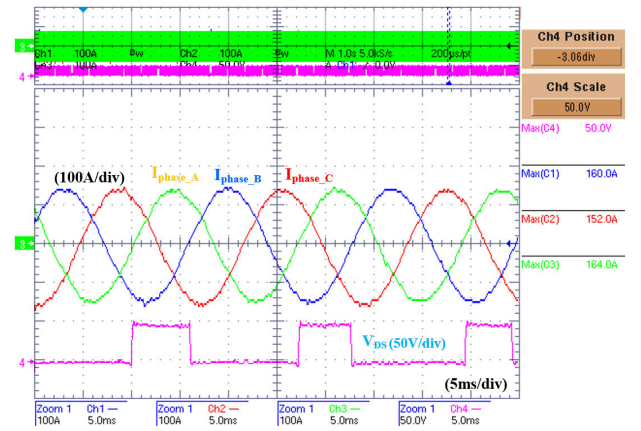


Fig. 25. Experimental waveforms of three-phase inverter setup at 20 kHz to 5 kW condition.



Fig. 26. Thermal image of the power module as evaluated in three-phase system at 20 kHz to 5 kW condition.

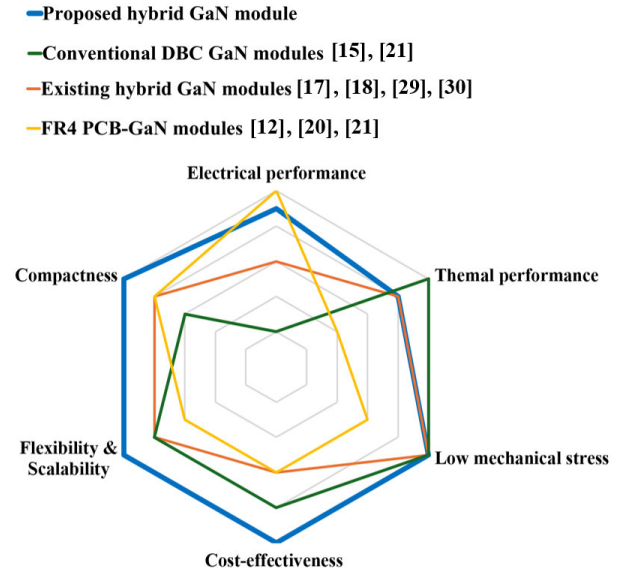


Fig. 27. Radar chart comparing the performance of the proposed GaN power module with existing modules housing four or more parallel GaN chips.

- 1) To improve thermal performance and mechanical strength in future study, the IMS aluminum baseplate can be replaced with a DBC substrate. In addition, integrating the heat sink into a pin-fin base plate can be an alternative technique to enhance heat transfer

capability. This proposed packaging concept is also suitable for 650-V bottom-cooled GaN chips such as GS66516B. To ensure the electrical clearance in the high-voltage power module without increasing power loop inductance, a thin FR4 fiberglass sheet of 0.15 mm can be inserted between the ac switching node and decoupling capacitor PCBs.

- 2) Thermomechanical simulation, power cycling, and thermal cycling tests might be performed to assess the reliability of the proposed power module under different temperature profiles.
- 3) Eventually, the entire inverter system with closed-loop control should be evaluated on the motor test bench using the standard driving cycles, which is planned for the near future.

VIII. CONCLUSION

This article presents the design and implementation of a compact GaN-based 360 A/100-V half-bridge module with dimensions of $44 \times 79 \times 10$ mm. By combining the advantages of existing power packaging solutions and addressing their limitations, the proposed structure achieves a uniform power and gate loop with low inductance, high thermal conductivity, robust switching waveforms, no mechanical stress on the devices, reduced complexity, cost-effectiveness, and expandable power capability. The DPT successfully validated the dynamic switching performance of the proposed power module, achieving 328 A for a single module with a dV/dt of 3.2 V/ns and a peak voltage of 60.8 V. In parallel configuration, a dual-module setup reached 400 A while maintaining the same dV/dt value and a maximum voltage of 49.6 V. In addition, it is configured as a 48-/24-V 3-kW buck converter and a 48-V 5-kW inverter to assess its thermal performance during steady-state operation. With a terminal arrangement similar to standard commercial products, this design is also compatible with both IMS and DBC packaging techniques, making it an attractive solution for a wide range of high-power automotive inverter systems.

APPENDIX

A. Gate Driver ICs Selection

In scenarios with high switching speed and frequency demands, the Si8271 gate driver IC can be replaced by the 1EDB7275F gate driver IC from Infineon, which offers notably higher source/sink currents of 5/9 A, as detailed in Table IX. In the proposed hybrid power module configuration, a separate gate driver board from the main power substrate provides flexibility for different applications. Due to the same footprint package and output pin configuration, the proposed gate driver board layout using the Si8271 can be slightly modified to accommodate the 1EDB7275F.

B. Experimental Extraction of Power Loop Stray Inductance

As depicted in Fig. 28, this testing setup aims to estimate the stray inductance of the power loop through the high-frequency oscillation of the V_{DS} waveforms. The power module is

TABLE IX
GATE DRIVER ICs SOLUTION FOR PROPOSED GATE DRIVER BOARD

Gate driver ICs	Output voltage	Input voltage	Source/sink current	Footprint package
Si8271GB (This work)	4 V - 30 V (isolated)	2.5 V-5.5 V	1.8 A / 4 A	SOIC-8
1EDBx275F (Optional)	4.5V - 20 V (isolated)	3 V- 15 V	5 A / 9 A	SOIC-8

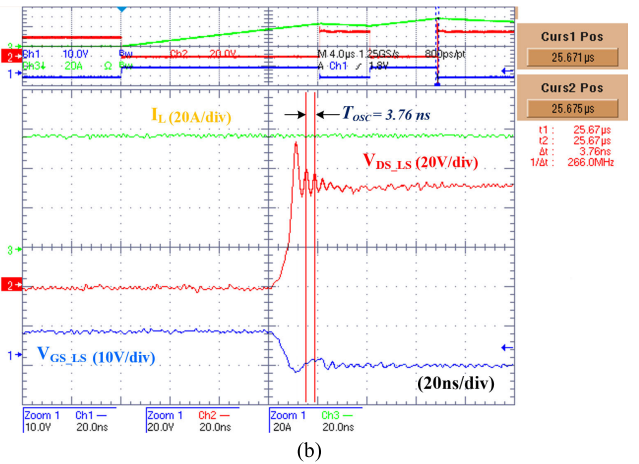
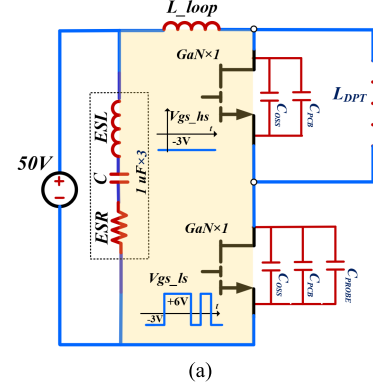


Fig. 28. Experimental setup for extracting power loop inductance. (a) Circuit diagram including parasitic components. (b) Oscillating waveforms resulting from resonant parasitic components.

soldered with one pair of high-side and low-side GaN devices. In addition, the attachment of the V_{DS} probe in the measurement loop is minimized to mitigate stray inductance effects. The DPT is tested at 50 V at 56 A using a 500-MHz bandwidth passive probe (Keysight-N2873A). During the turn-off transient period, the power loop inductance (L_{loop}) and the equivalent series inductance (ESL) of decoupling capacitors resonate with the output capacitance (C_{oss}) of the bottom-side GaN device, the PCB parasitic capacitance (C_{pcb}), and the output capacitance of the passive probe (C_{probe}). Fig. 28(b) shows that the oscillation period (T_{osc}) is about 3.76 ns, equivalent to a ringing frequency of 266 MHz. There are three ceramic decoupling capacitors (1 μ F-100 V) with the part number GCM21BC72A105KE35.

The self-resonant frequency (SRF) of the capacitor can be determined by plotting the impedance (S_{11} parameter) using the S2P file format provided by the Murata manufacturer, as shown in Fig. 29. The SRF value of the

TABLE X

COMPARISON OF SWITCHING SPEEDS BETWEEN PROPOSED GaN POWER MODULE AND Si POWER MODULES IN 48 V MOTOR DRIVE APPLICATIONS

Ref	N° of parallel devices	Device packages	Switching speed (dV/dt)	V_{DC} / peak voltage	Testing methods	Applications	Switching frequencies
[3]	3×Si MOSFETs	POWERSFN	0.15 V/ns	48 V / 50 V	Inverter - 5 kW	Motor drive	12 kHz
[46]	4×Si MOSFETs	Bare die	~0.1 V/ns	48 V / 73 V	DPT - 300 A	Motor drive	10 kHz
[47]	4×Si MOSFETs	H ² PAK-6	~0.25 V/ns	48 V / 59.1 V	Inverter - 4.5 kW	Motor drive	12 kHz
[48]	4×Si MOSFETs	TOLT	~0.18 V/ns	48 V / 105 V	DPT - 150 A	Motor drive	N/A
[49]	Si MOSFETs	Bare die	~0.17 V/ns	48 V / 57.6 V	DPT - 200 A	Motor drive	8 kHz
[50]	4×Si MOSFETs	Bare die	~0.48 V/ns	52 V / 64 V	DPT - 370 A	Motor drive	8 kHz
This work	4×GaN	GaNpx (near chip scale)	3 V/ns	48 V / 49.6 V	DPT - 200 A	Motor drive	20 kHz
			3 V/ns	48 V / 60.8 V	DPT - 328 A		
				48 V / 50 V	Inverter - 5 kW		

GCM21BC72A105KE35 capacitor is 6.65 MHz, resulting in an ESL value of 0.57 nH ($ESL = 1/(4\pi^2 SRF^2 C)$). Subsequently, the power loop inductance value is approximated in the following equation:

$$\begin{aligned}
 L_{loop} &\approx \frac{T_{osc}^2}{4\pi^2(C_{oss} + C_{pcb} + C_{probe})} - \frac{ESL_{decoupl_cap}}{N_{decoupl_cap}} \\
 &\approx \frac{3.76 \text{ ns}^2}{4\pi^2(250 \text{ pF} + 2 \text{ pF} + 10 \text{ pF})} - \frac{0.57 \text{ nH}}{3} \\
 &\approx 1.175 \text{ nH}
 \end{aligned} \quad (14)$$

where C_{oss} is the output capacitance of 250 pF obtained from the datasheet at 50 V, C_{pcb} is estimated as 2 pF, and the probe capacitance (Keysight-N2873A) ranges from 10 to 25 pF. Due to the use of three parallel decoupling capacitors ($N_{decoupl_cap} = 3$), the total ESL value of the decoupling capacitors is reduced from 0.57 to 0.19 nH. Therefore, the power loop inductance value is 1.175 nH, which is close to the simulation value of 1.14 nH.

C. Comparison of Switching Characteristics in 48-V Motor Drive Applications

Table X compares the experimental results of switching speeds, overshoot voltage, and switching frequency between the proposed GaN power module and other Si power modules in 48-V motor drive applications. Existing traction power modules have dV/dt values ranging from 0.15 to 0.48 V/ns, which is considerably less than the 3 V/ns of GaN power modules. In addition, the DPT and inverter test results indicate that the proposed GaN power module has the lowest overshoot voltage.

D. Mechanical Stress Analysis of Proposed GaN Power Module

As previously mentioned, compared to a multilayer FR4-PCB solution, the proposed GaN power module offers a rigid structure due to the stronger mechanical support provided by its metal baseplate and bottom-side cold plate attachment. This part of the article focuses on verifying the mechanical stress on the power module using ANSYS Static Structural simulation. The mechanical simulation includes the application of standard earth gravity in the $-Z$ direction and frictionless support at the bottom to represent the connection between the power module and the cold plate. A preload force of 1697.6 N has

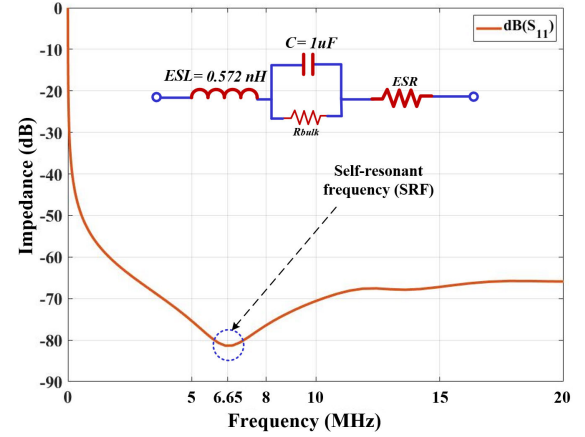


Fig. 29. Impedance versus frequency plot of the GCM21BC72A105KE35 capacitor.

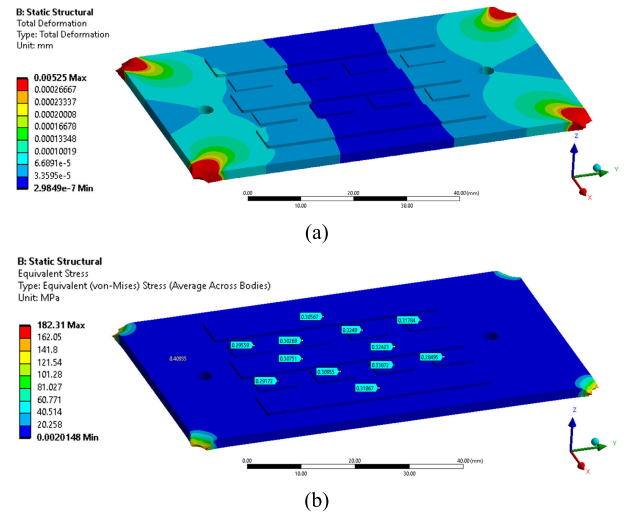


Fig. 30. Mechanical stress simulation results. (a) Total deformation. (b) Equivalent stress (the observed deformation is amplified by a factor of 320).

been applied for M3 hex socket cap screws in A2 304 quality, which are located at the four corners of the power module. The material parameters used in the mechanical simulation are provided in Table XI.

As illustrated in Fig. 30, the mechanical stress distribution across the GaN chips is uniform, with the maximum deformation occurring at the four corners of the baseplate, measuring 5.25×10^{-3} mm. The equivalent stress is concentrated at these

TABLE XI
MECHANICAL PROPERTIES OF MATERIALS

Materials	Elastic modulus (GPa)	Poisson's ratio	Density (g cm ⁻³)
Aluminum (Baseplate)	68.9	0.33	2.69
Copper Alloy (Copper bars)	110	0.35	8.3
GaN-on Si (GaN chips)	160	0.26	6.15
SAC305 (Solder paste)	43	0.36	7.38

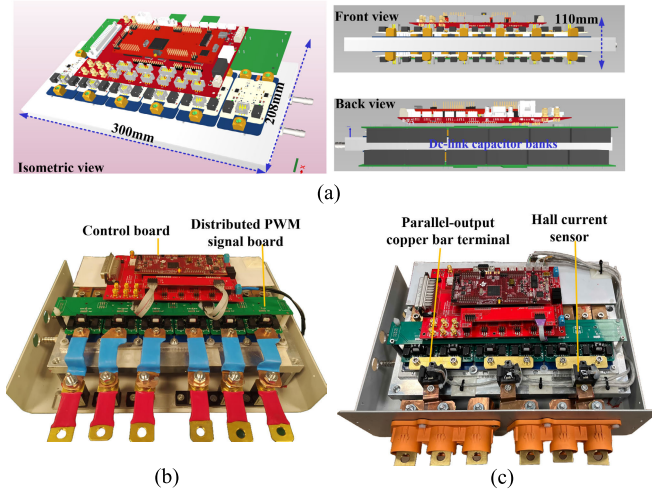


Fig. 31. GaN-based inverter prototypes. (a) 3-D CAD model from different perspectives. (b) 12-kW six-phase inverter configuration. (c) 25-kW six-phase inverter configuration.

corners due to the force applied by the tightening screws, with a peak value of 182 MPa. This stress level remains below the yield stress of Al6061, which is 276 MPa. The GaN chips, meanwhile, experience negligible mechanical stress, with a Von Mises stress value of 3×10^{-1} MPa. Future study might investigate reliability concerns due to thermomechanical stress caused by differences in the coefficients of thermal expansion (CTE) between materials, which can be validated through the thermal cycle testing setup.

E. Prototype Development of a High-Power GaN-Based Traction Inverter Using the Proposed Power Module

As illustrated in Fig. 31(a), either six power modules with one dc-link capacitor bank or 12 power modules with two dc-link capacitor banks can be arranged on one side or both sides of the liquid cold plate. This flexible arrangement supports various inverter topologies and allows scalability to different power levels. Fig. 31(b) shows a 12-kW six-phase inverter.

As depicted in Fig. 31(c), using two parallel power modules can increase the power to 25 kW. Alternatively, the configuration can be adapted to a 12-kW three-phase inverter with six power modules. Thus, depending on the number of phases of the motor and the required power, the inverter structure can be easily modified accordingly.

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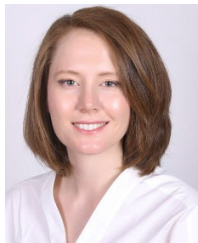
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